

TPMs and the Linux Kernel

Unlocking a better path to hardware security

Ignat Korchagin
@ignatkn

CLOUDFLARE

What is a TPM?

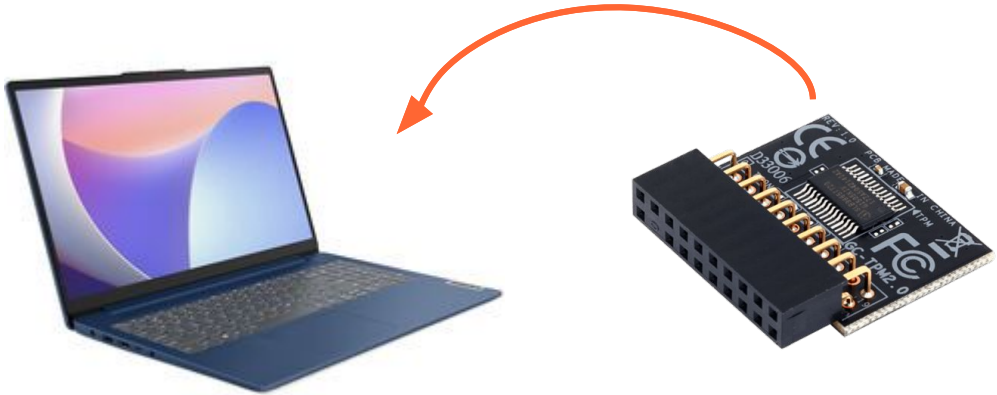
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What is a TPM?



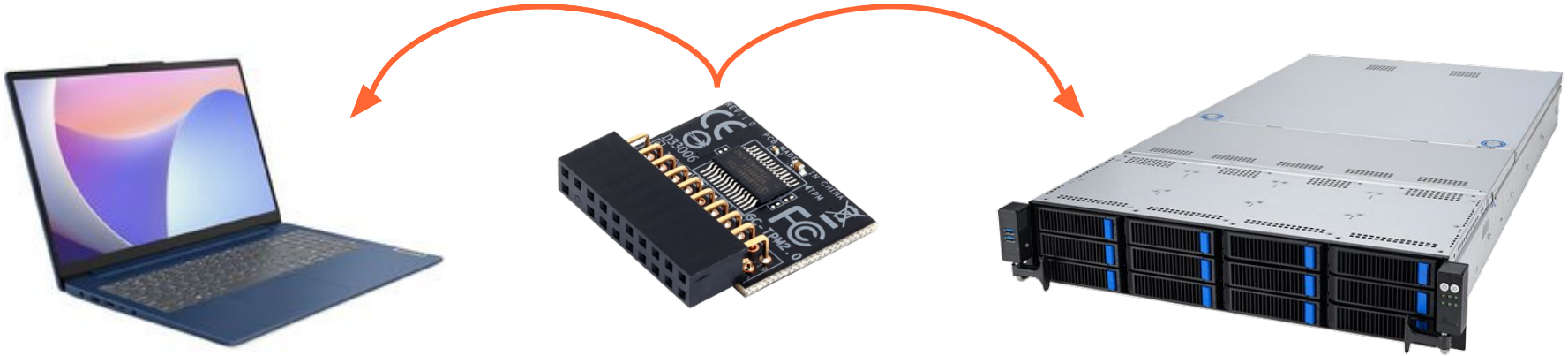
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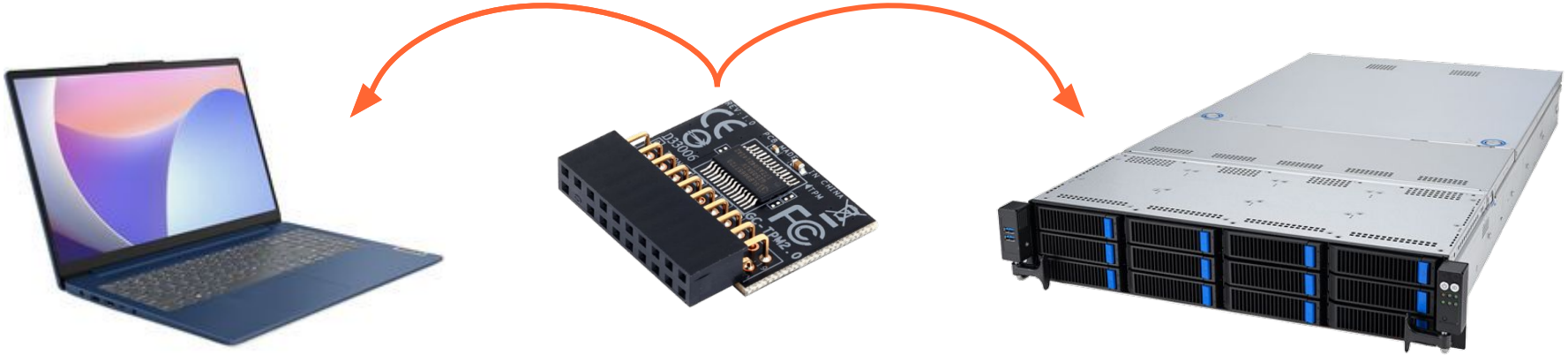
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What is a TPM?



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What is a TPM?



- A discrete security chip on modern laptops and servers
- Passive, non-intrusive: only responds to commands and performs cryptographic operations
- Foundation for platform integrity, authentication and remote attestation
- Can handle cryptographic keys

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This talk is not about system integrity or attestation

Can TPM help me with my keys/secrets?

Why so little open source software has TPM support?

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Application keys in the TPM



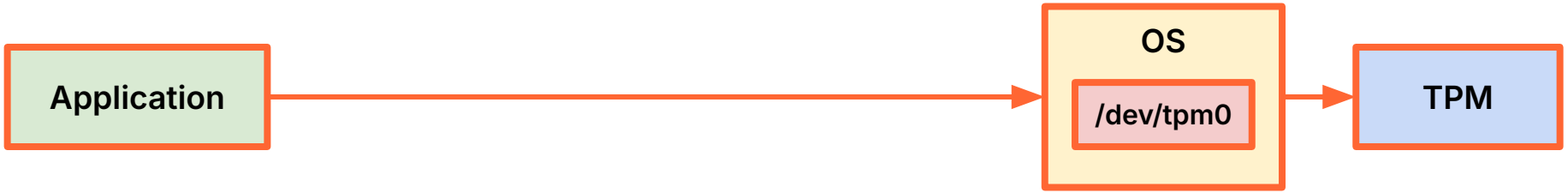
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Application keys in the TPM



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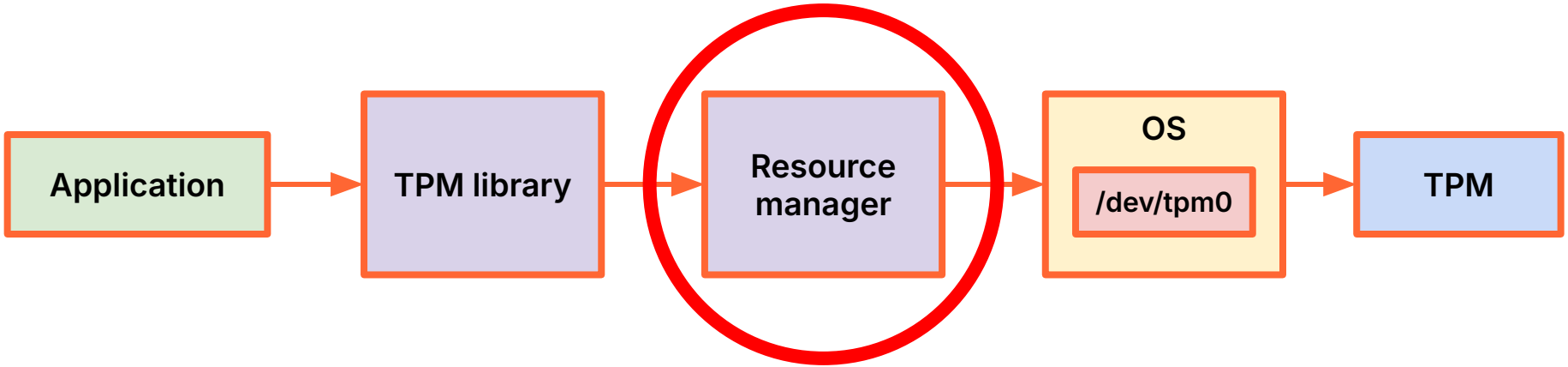
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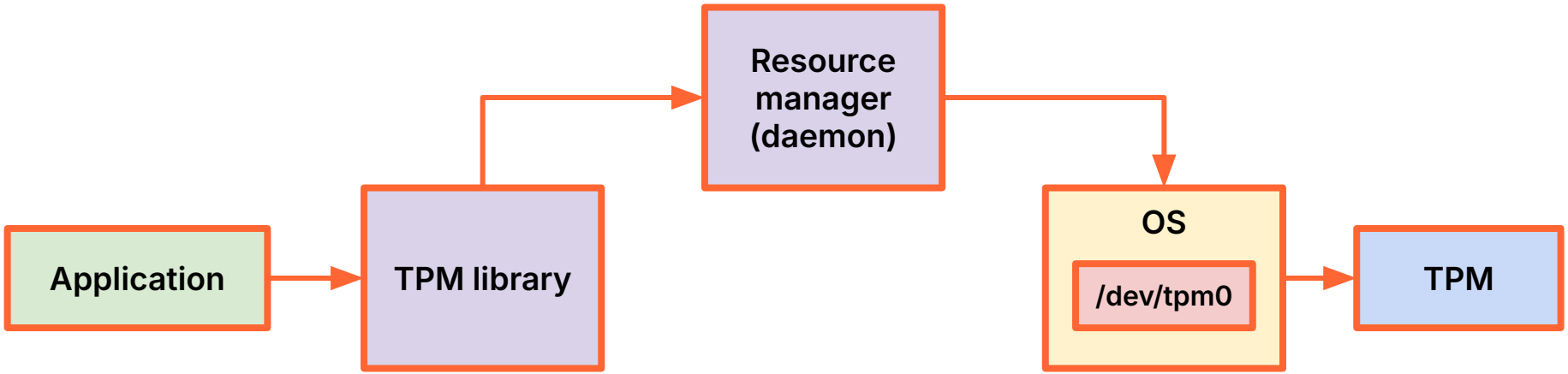
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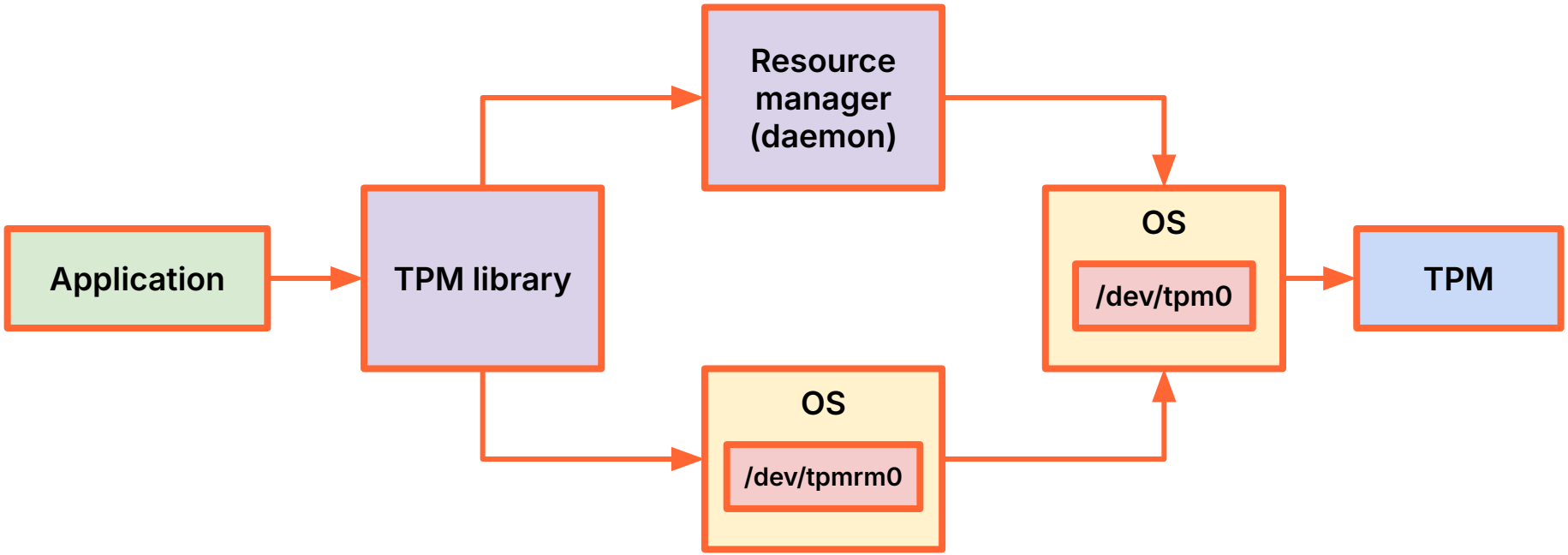
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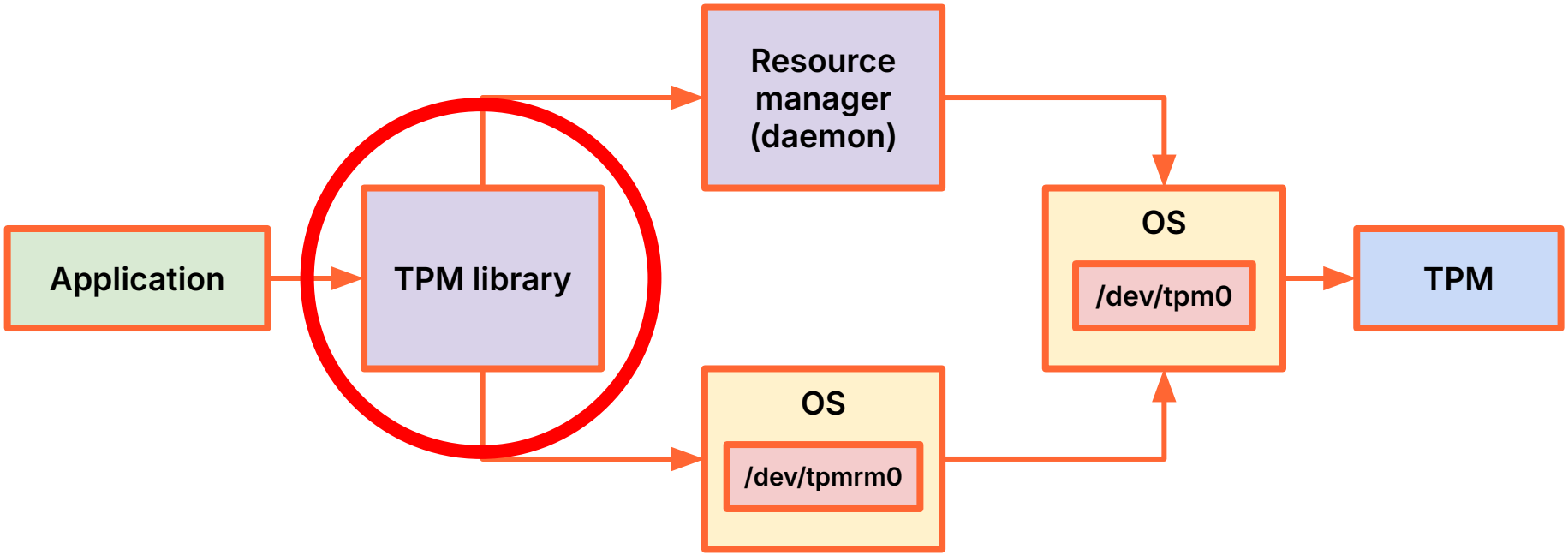
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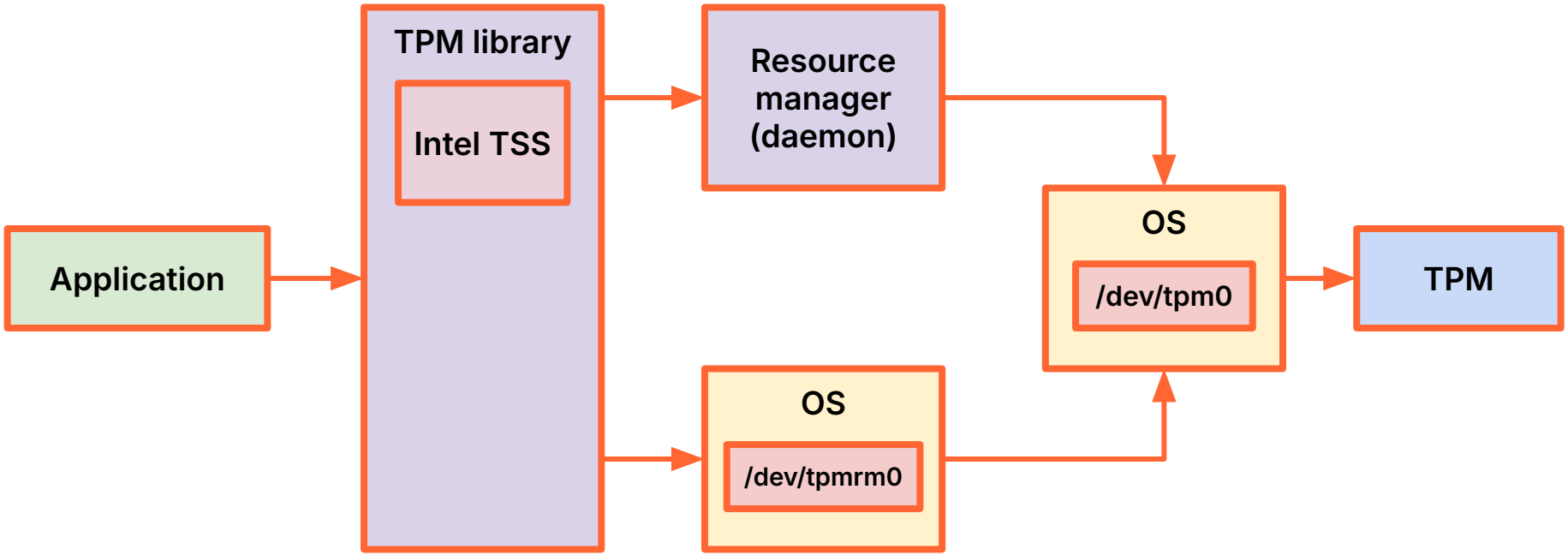
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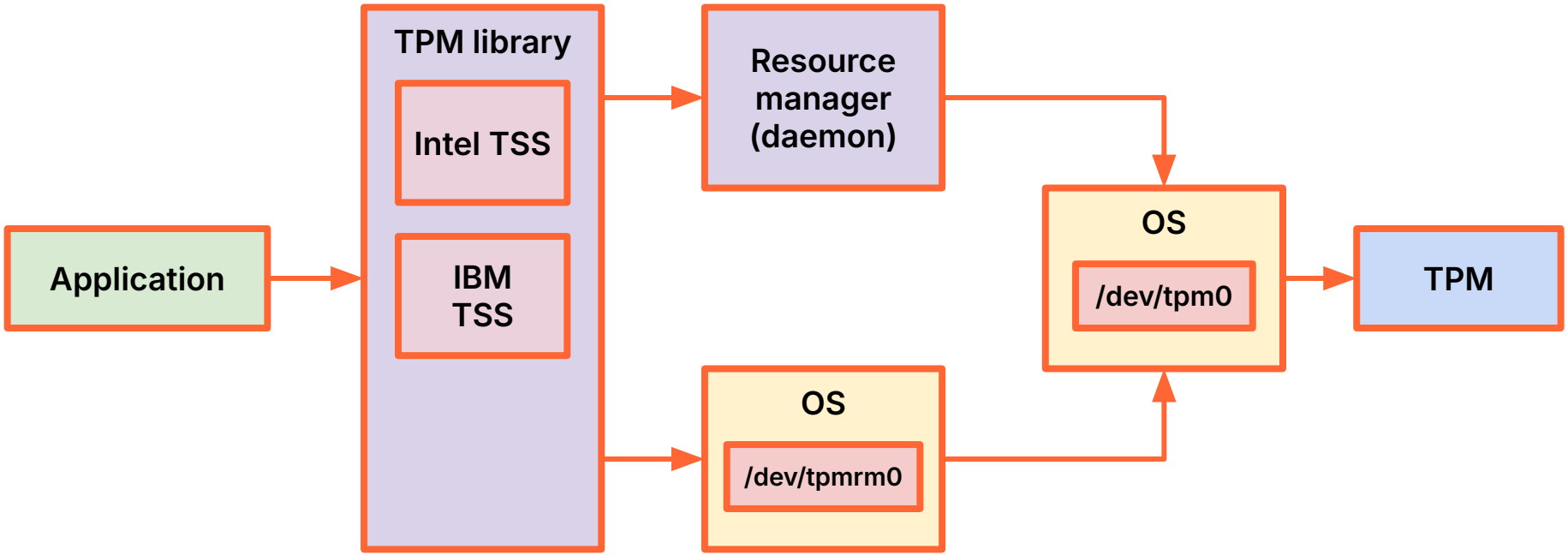
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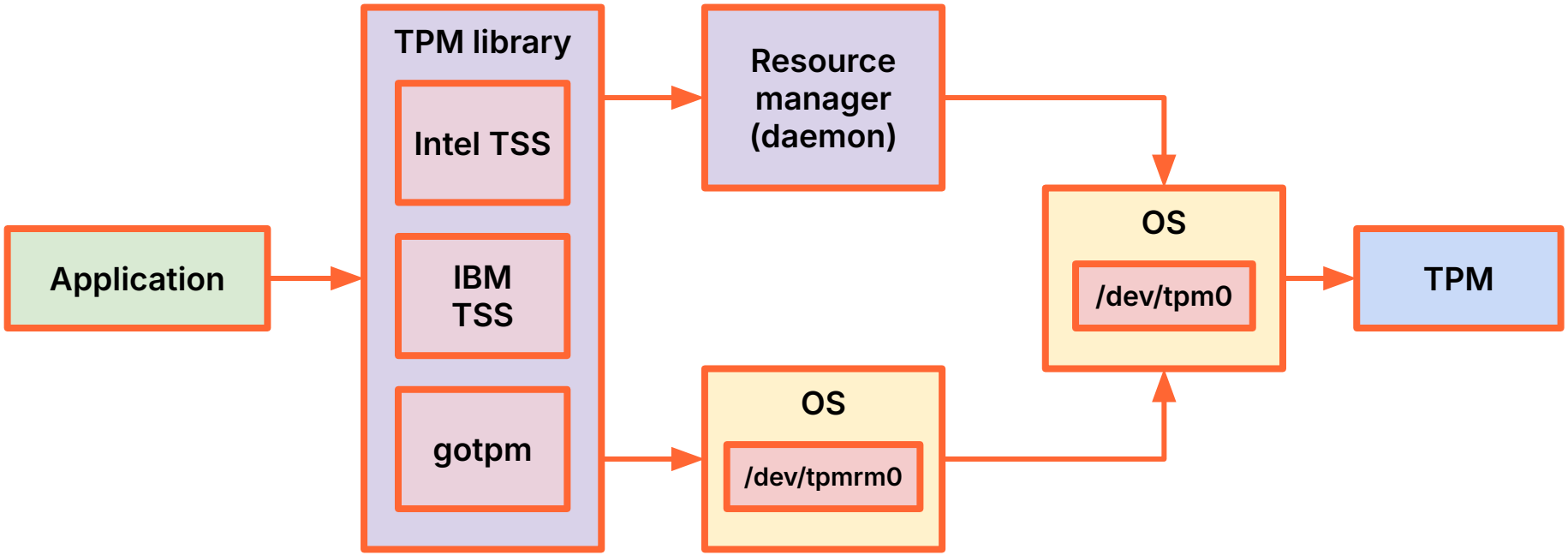
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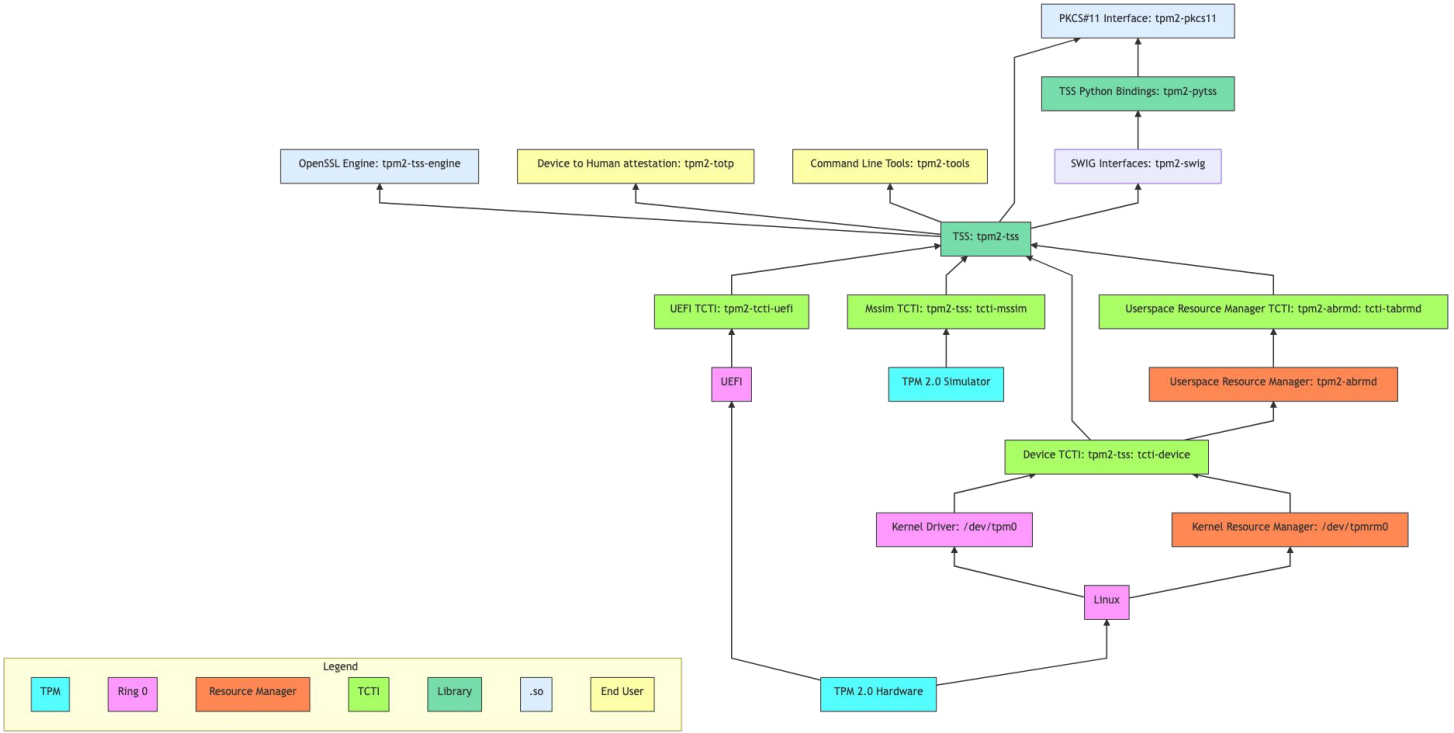
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Application keys in the TPM



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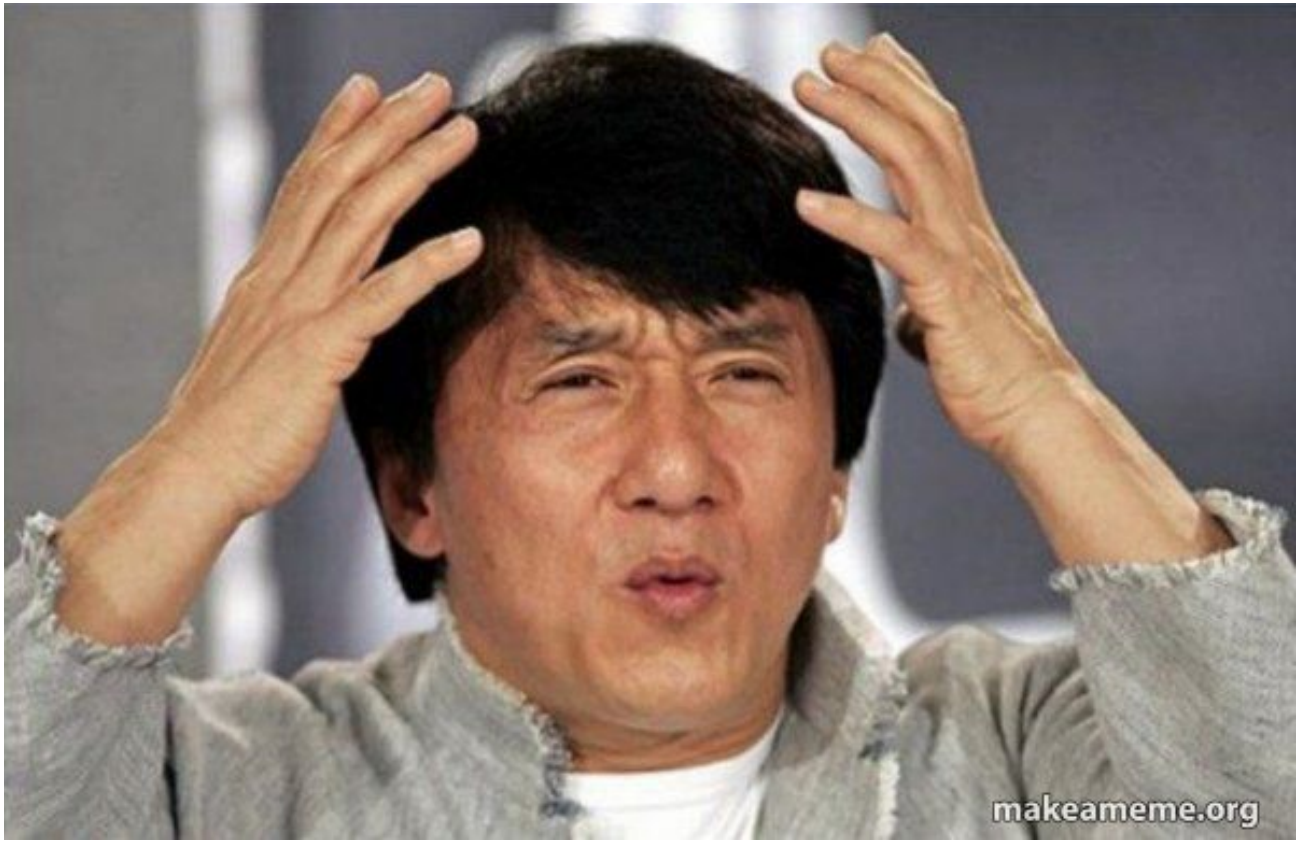
TPM2 software stack



https://tpm2-software.github.io/

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Application keys in the TPM



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Access permissions for the TPM

```
ignat@dev:~$ echo hello > /dev/tpmrm0
-bash: /dev/tpmrm0: Permission denied
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```

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Linux Kernel key retention service

AKA keyrings or keystore

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Linux Kernel key retention service

Application

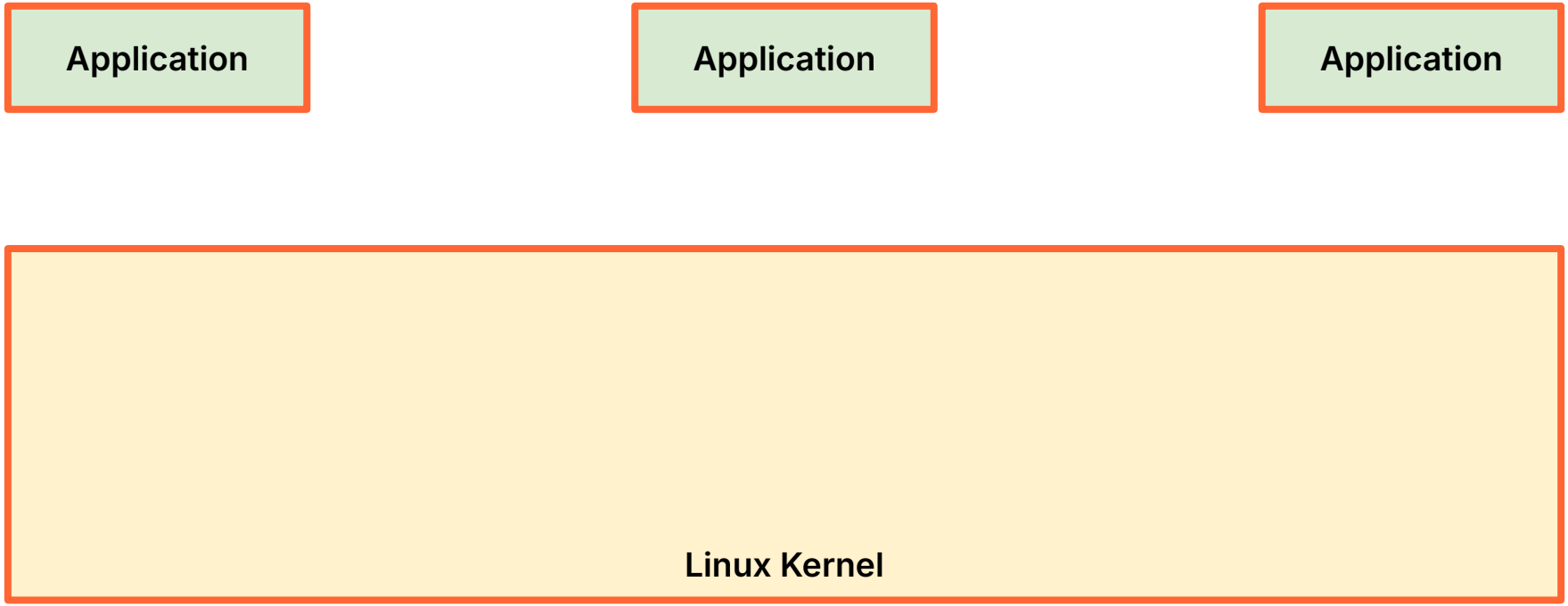
Application

Application

https://www.kernel.org/doc/html/latest/security/keys/core.html

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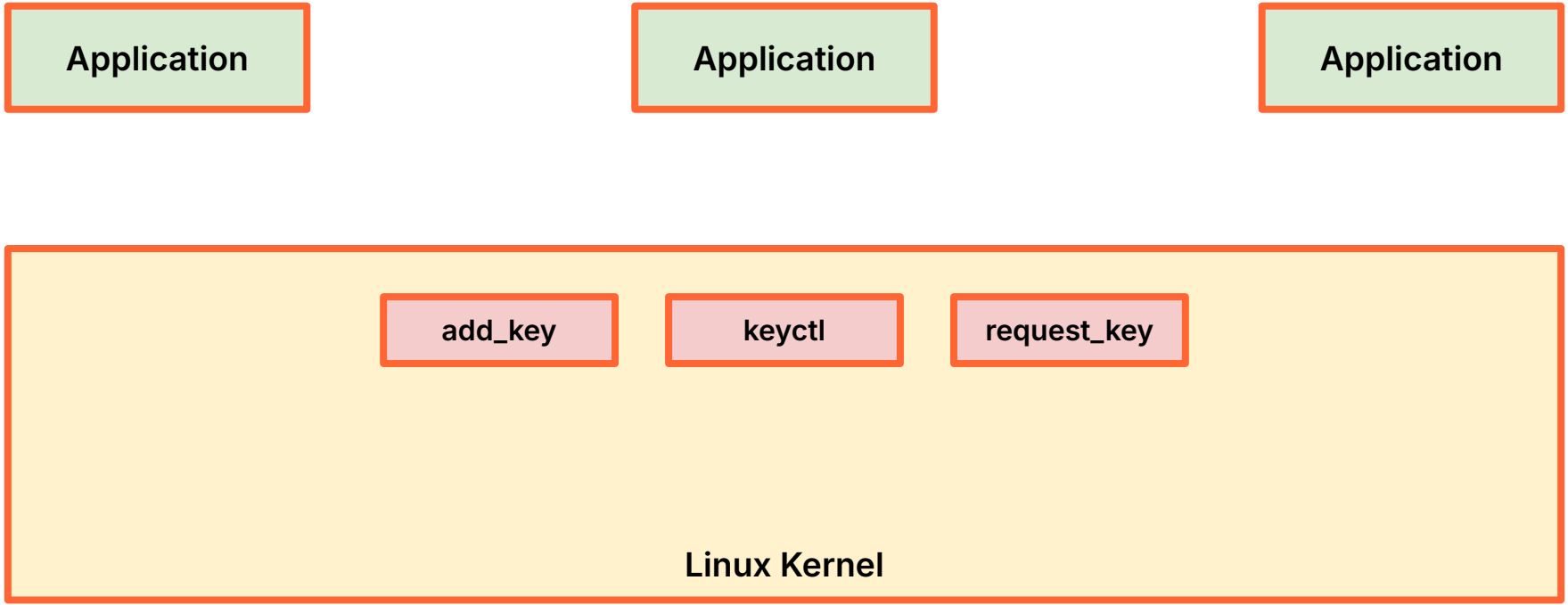
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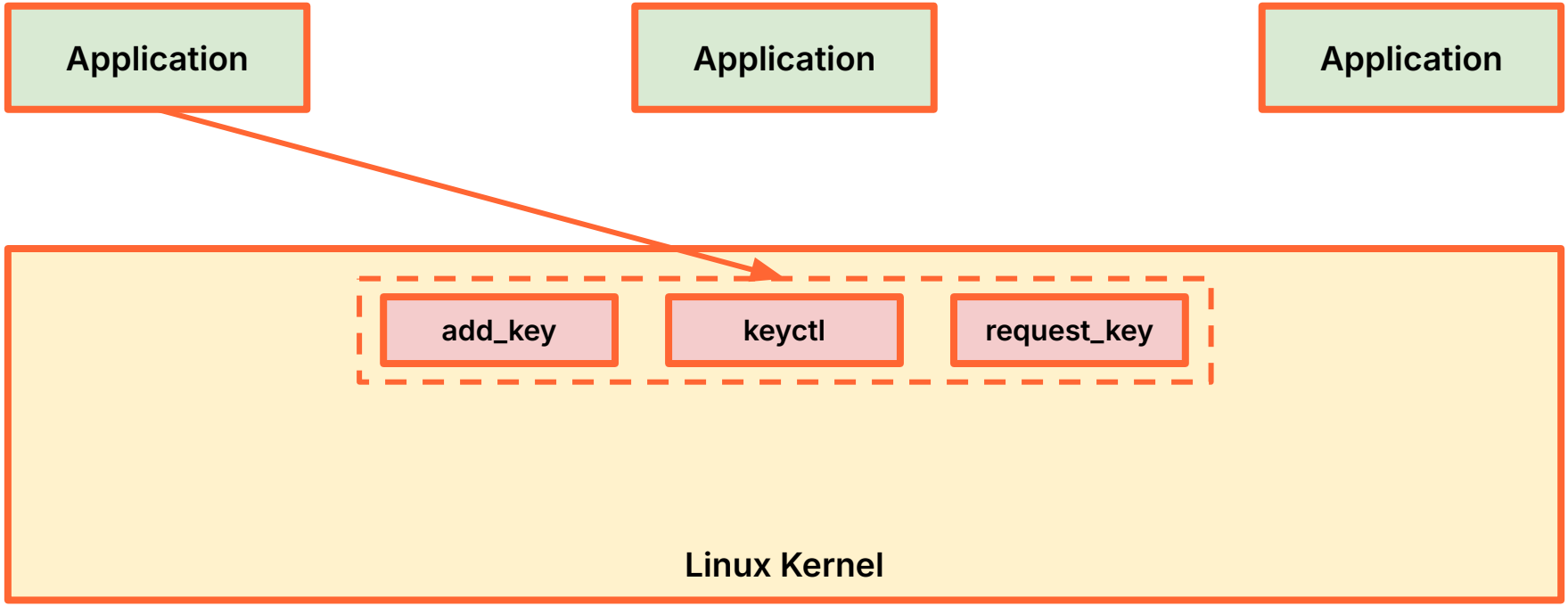
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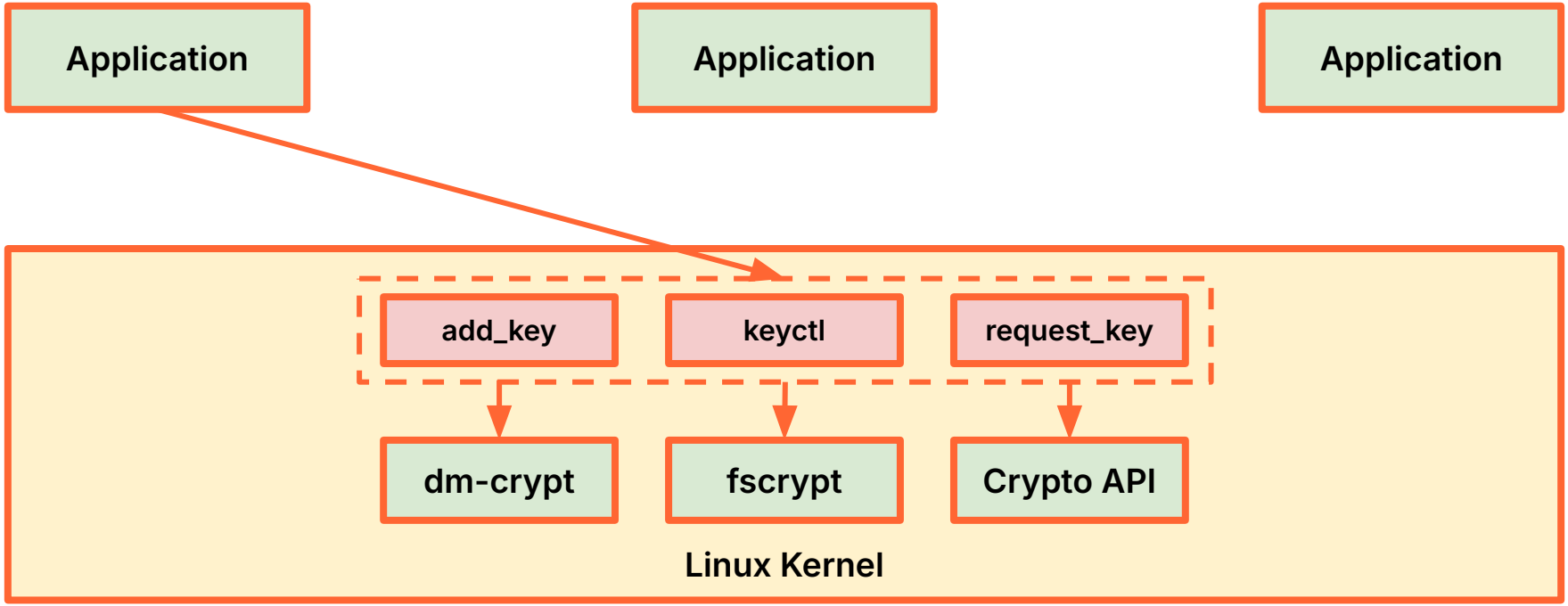
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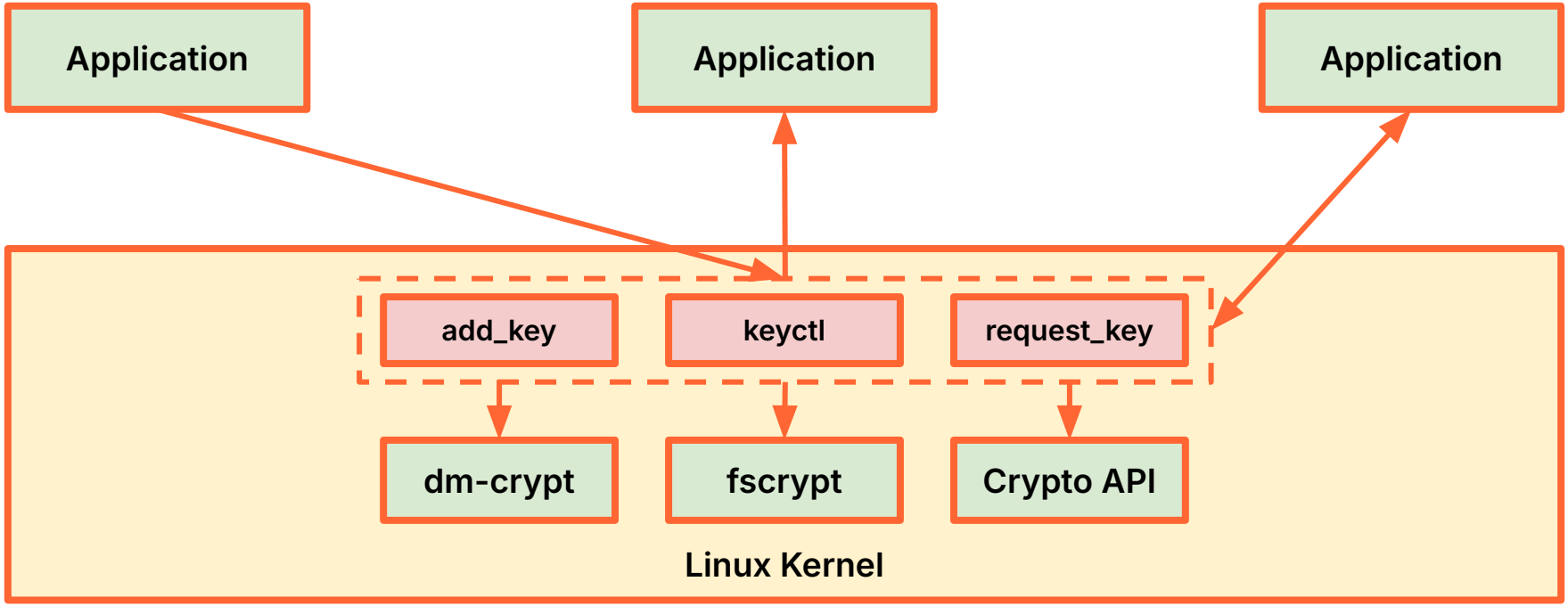
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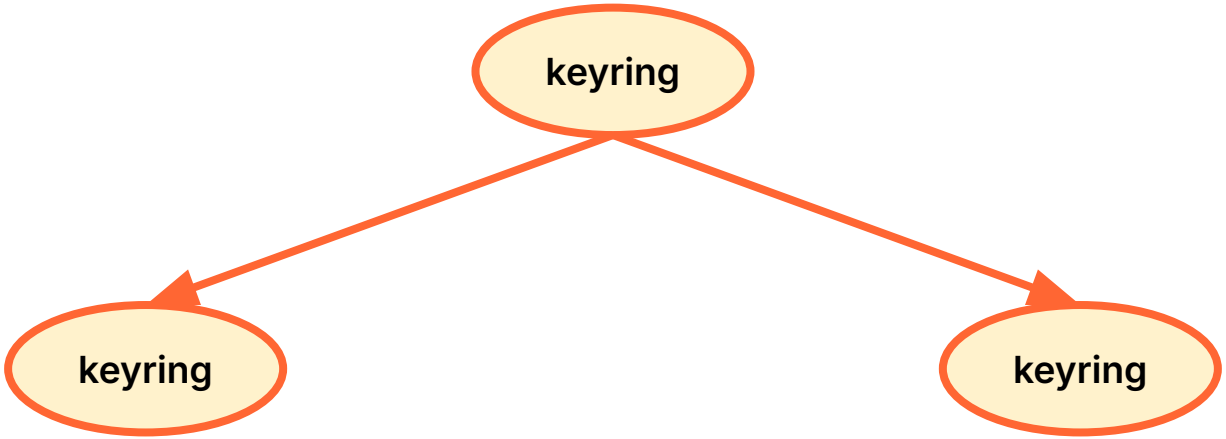
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Keys and keyrings



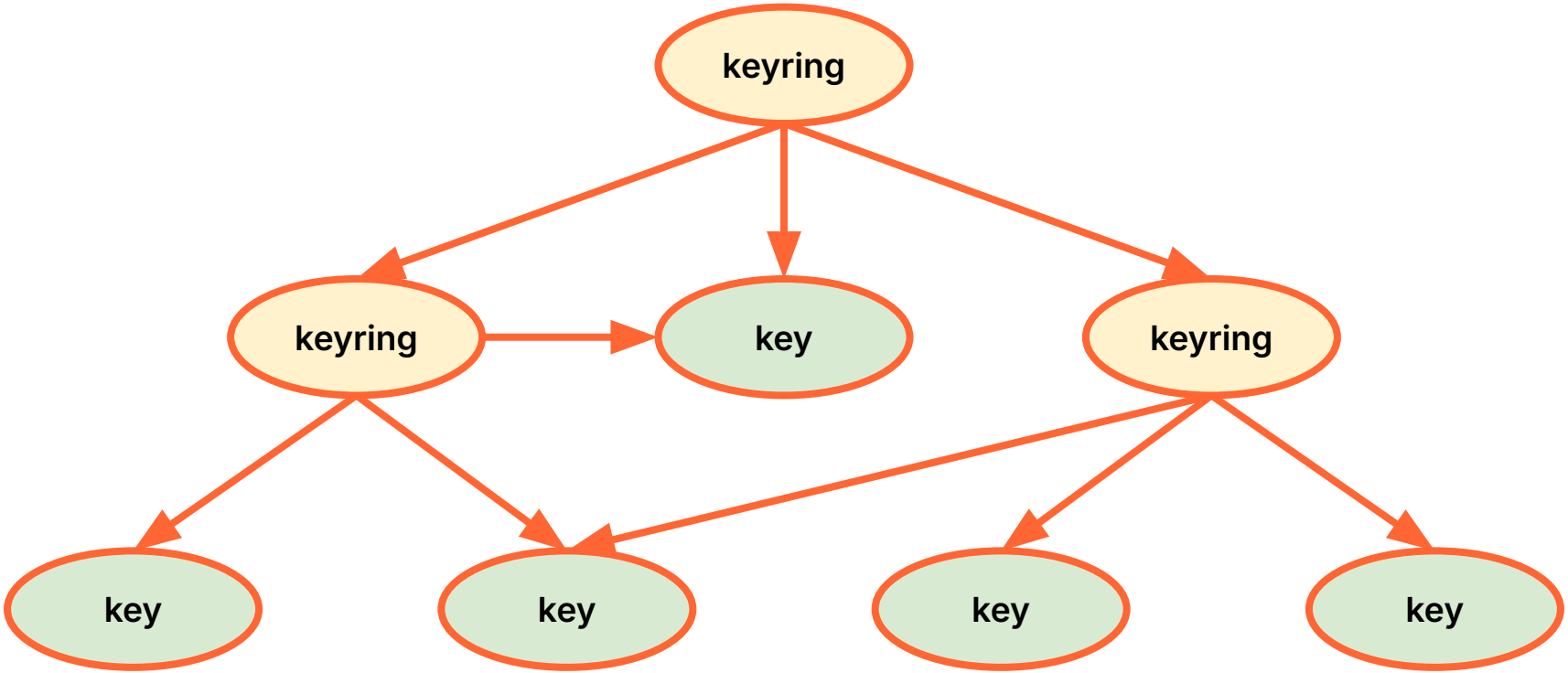
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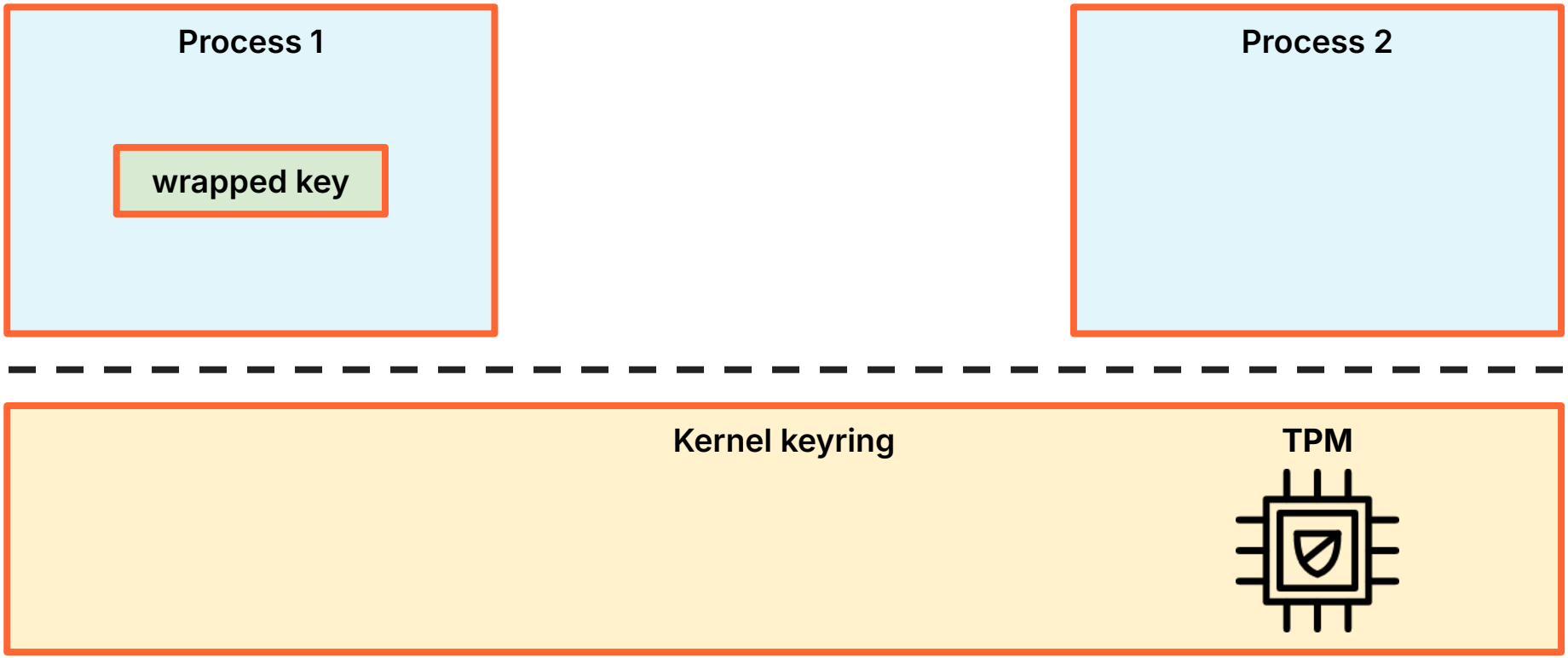
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Linux keystore and TPMs

Better together?

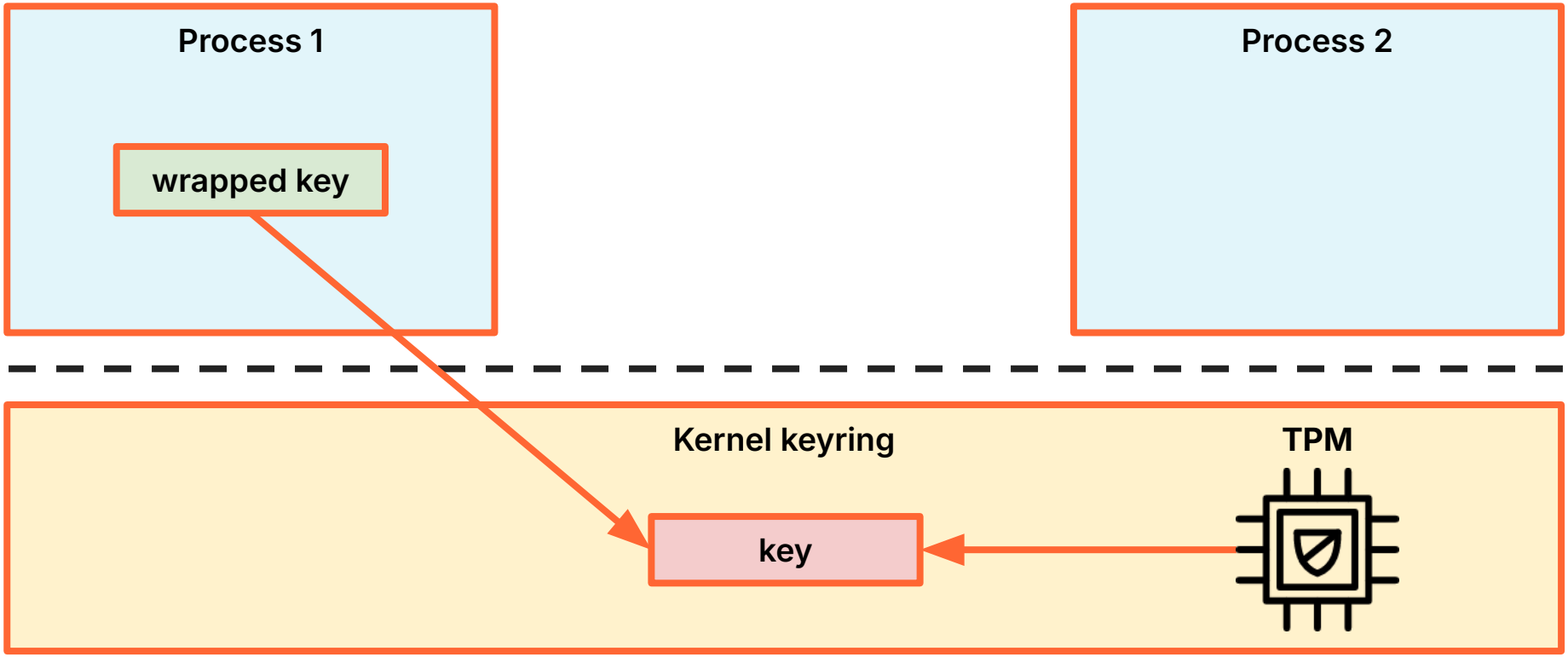
CLOUDFLARE

Trusted keys



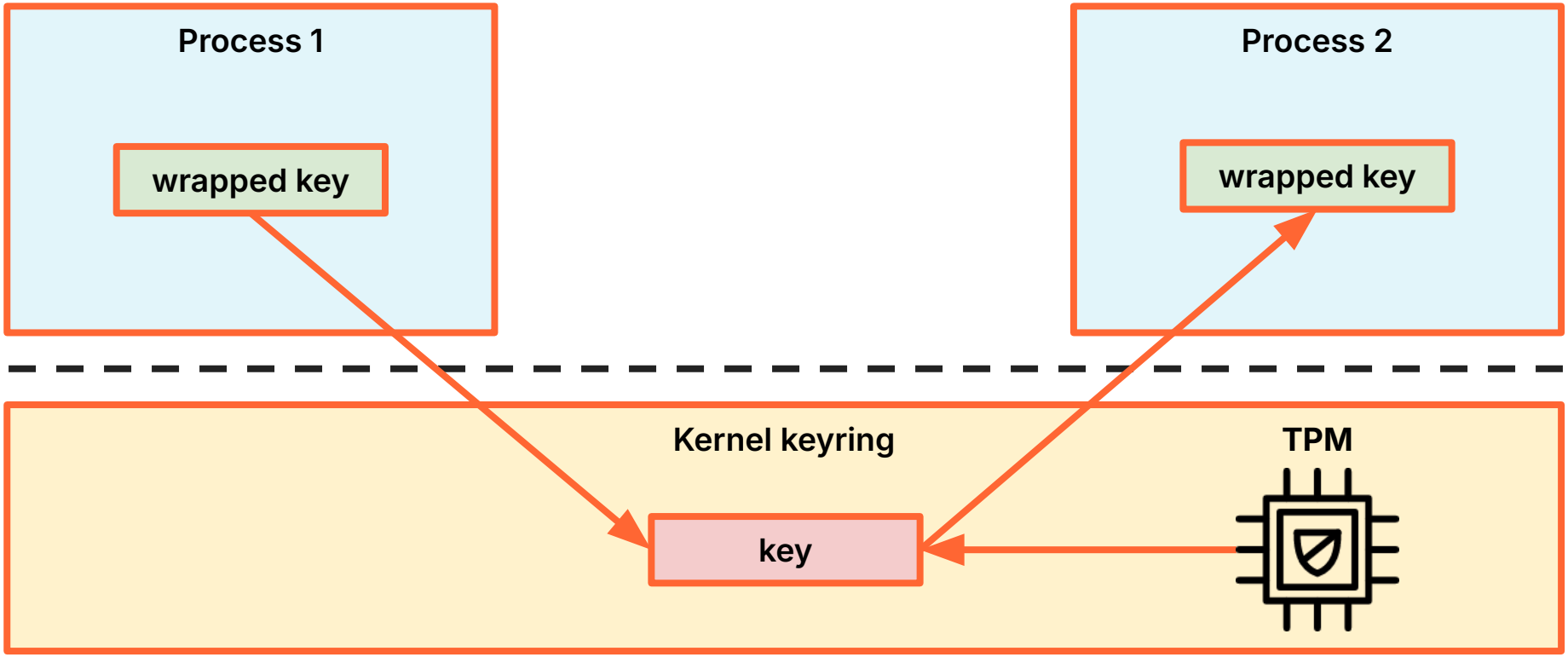
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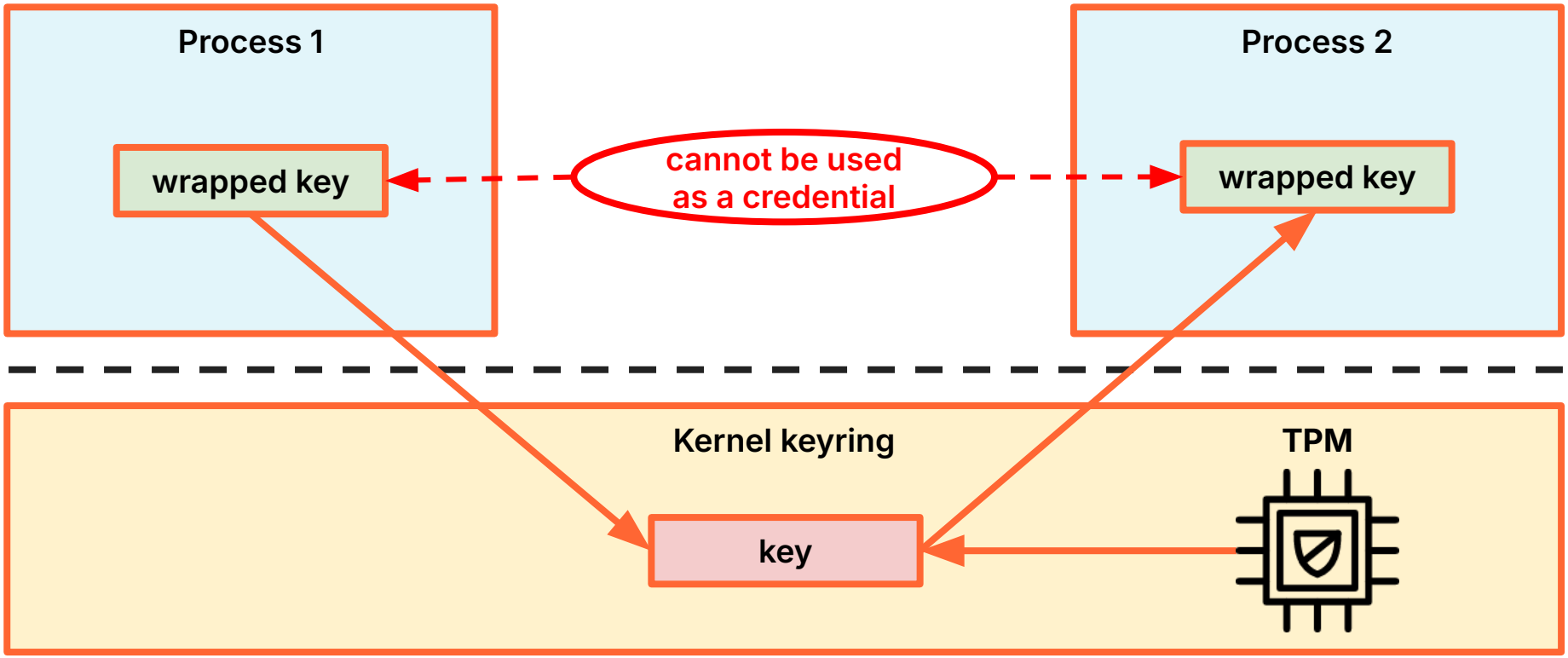
CLOUDFLARE

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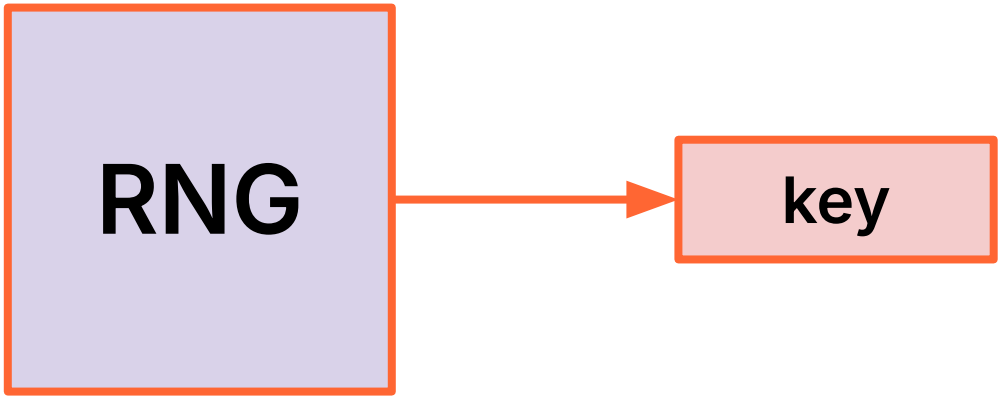
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Wrapped keys



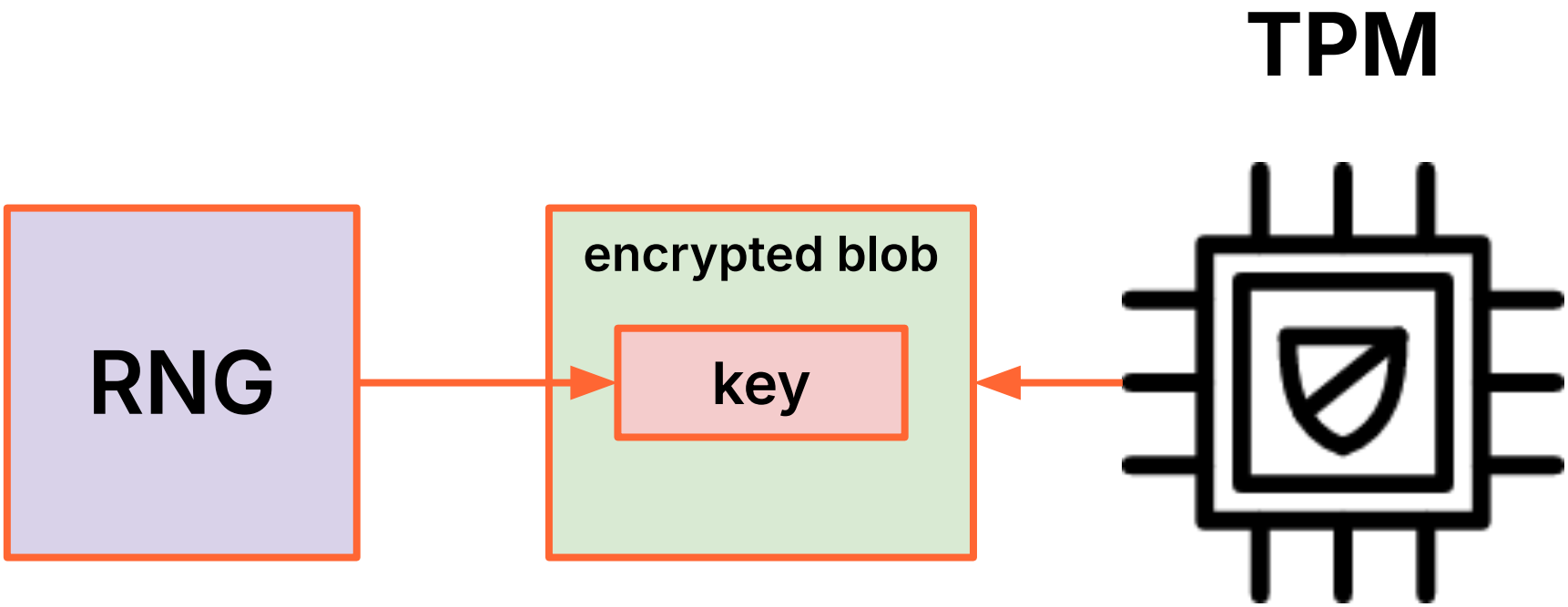
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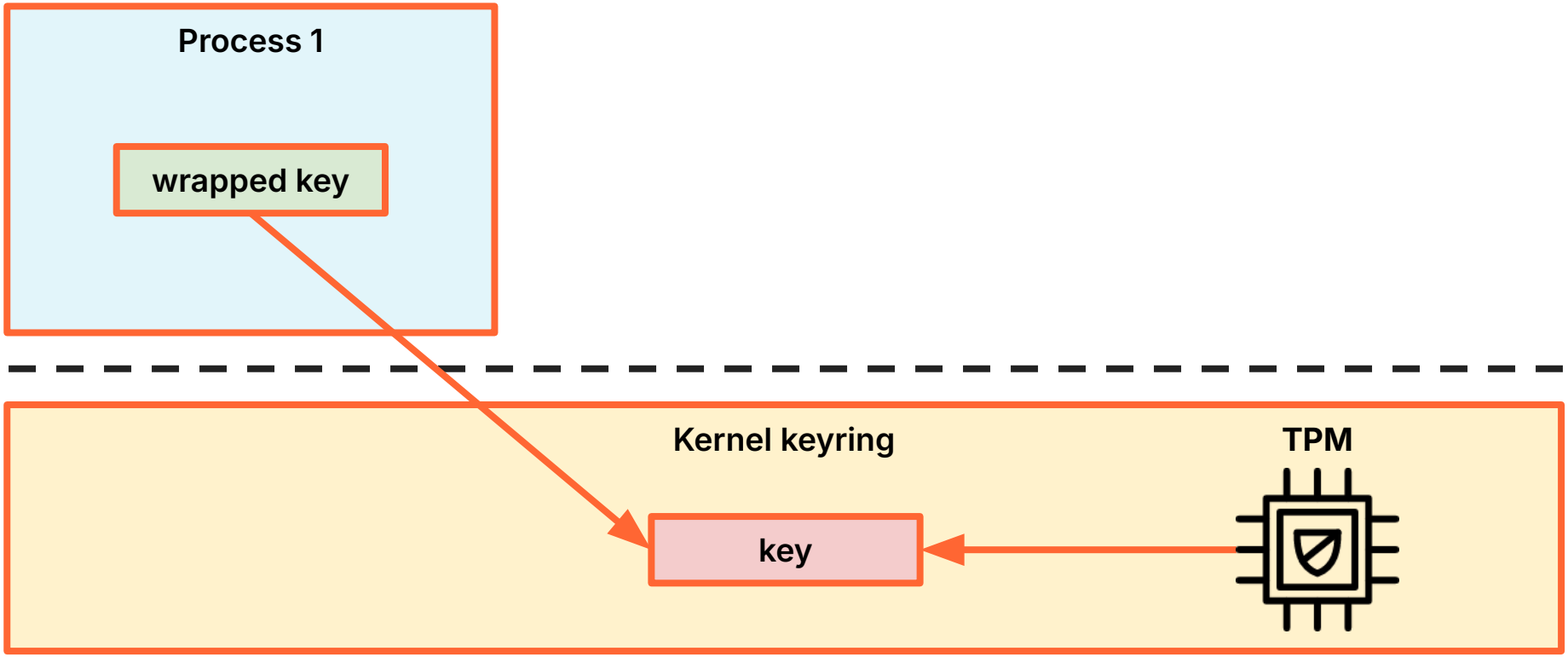
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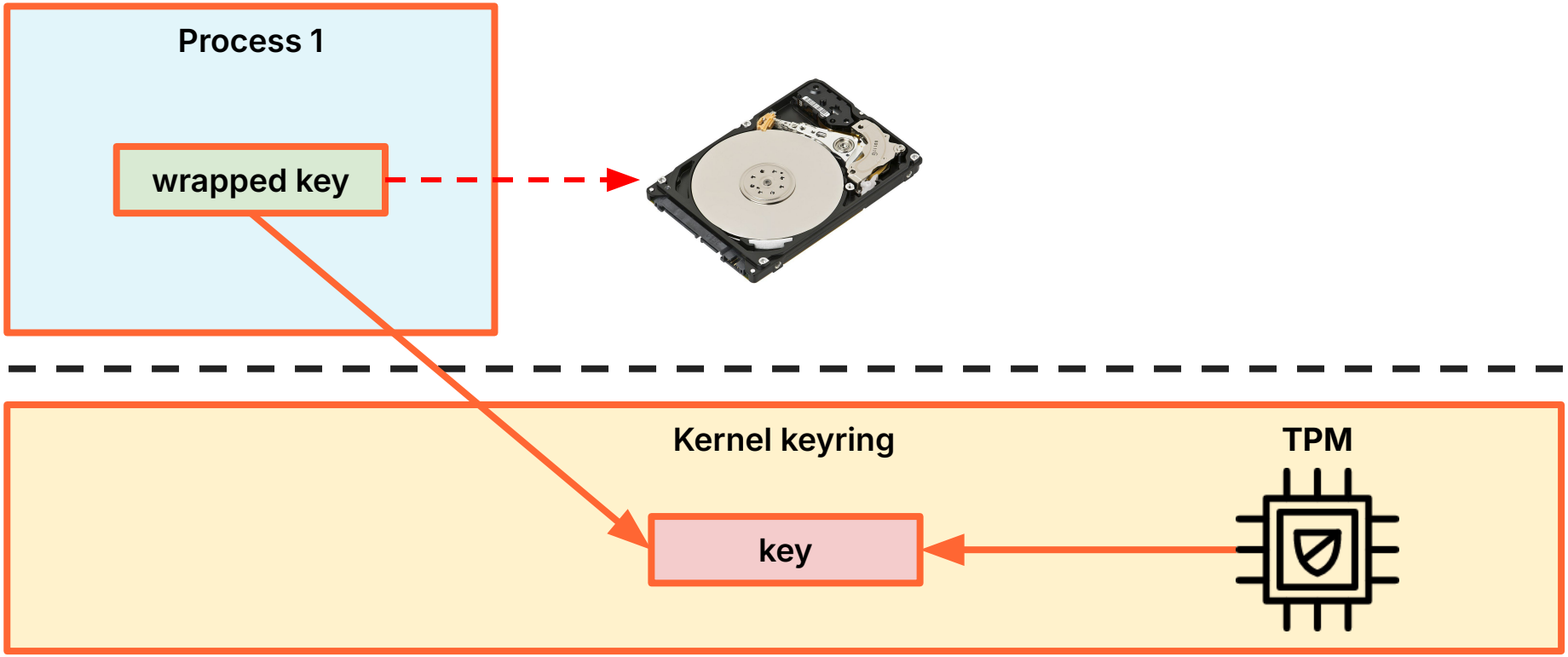
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Trusted key management



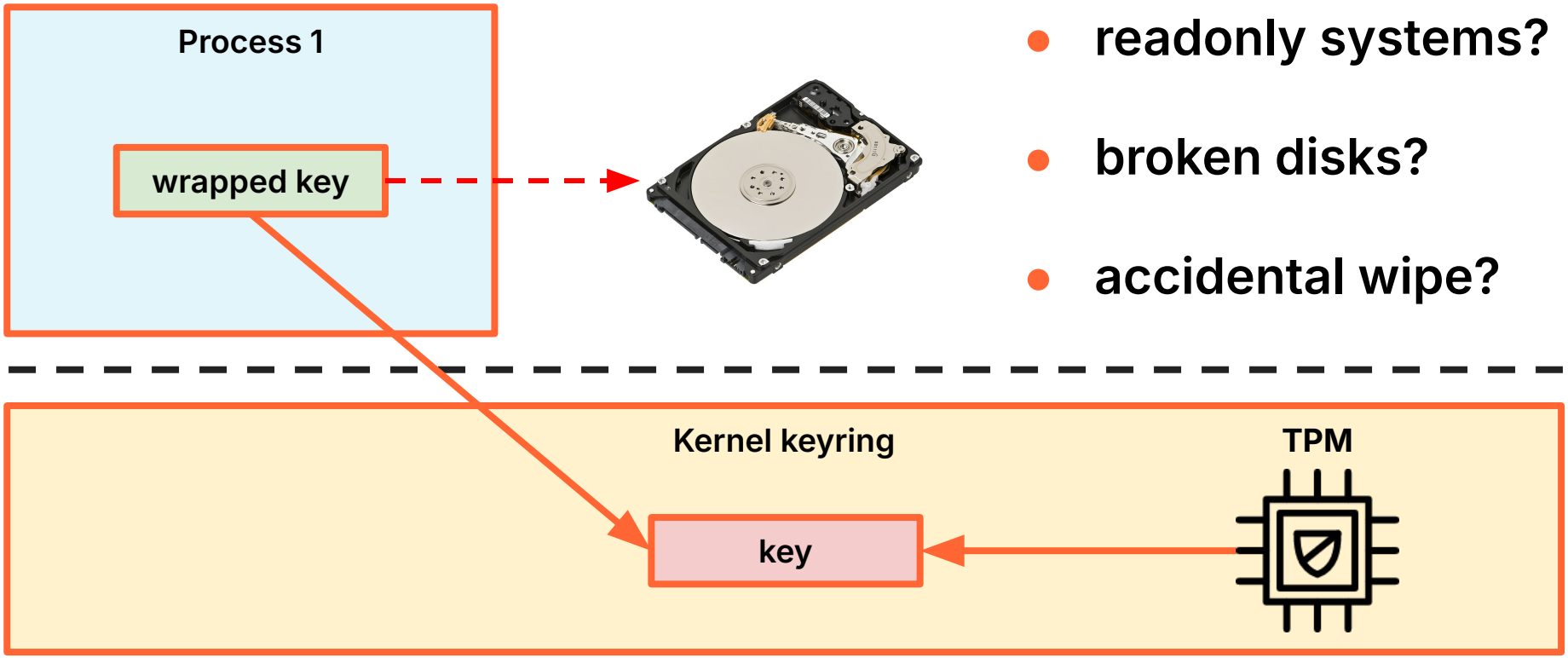
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Trusted key management



CLOUDFLARE

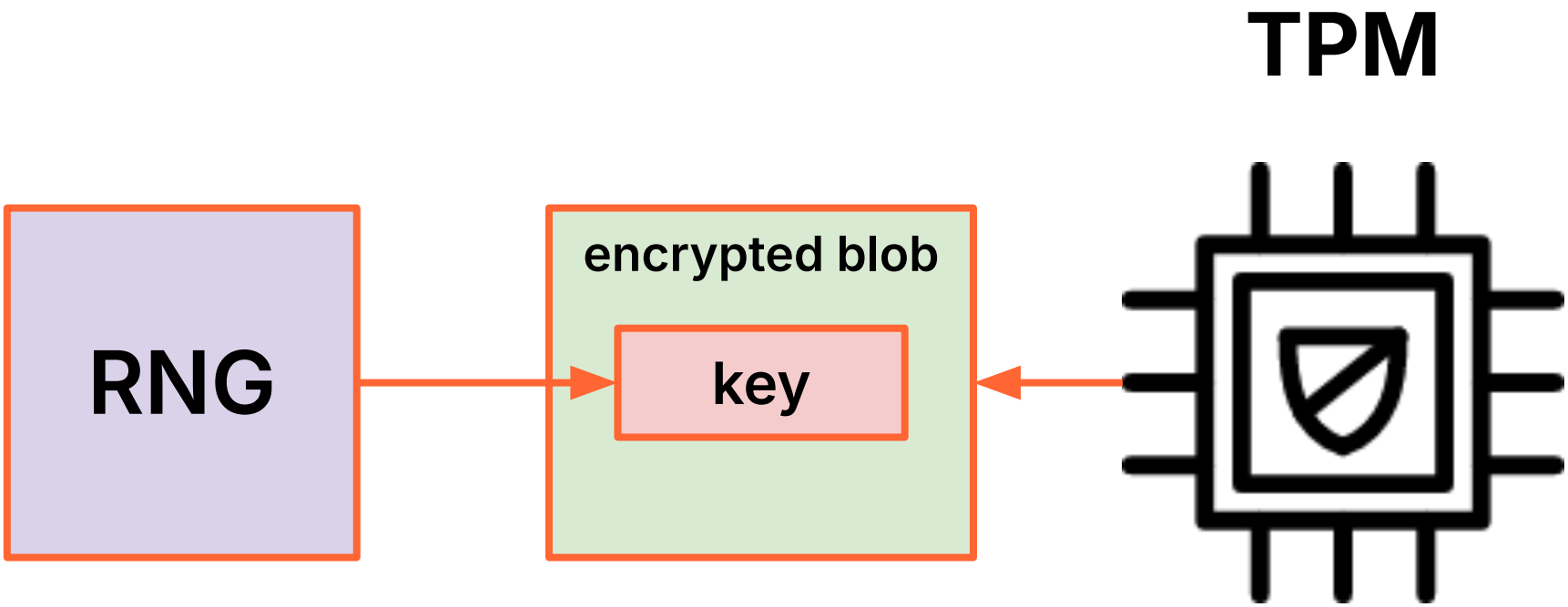
Trusted key management



- readonly systems?
- broken disks?
- accidental wipe?

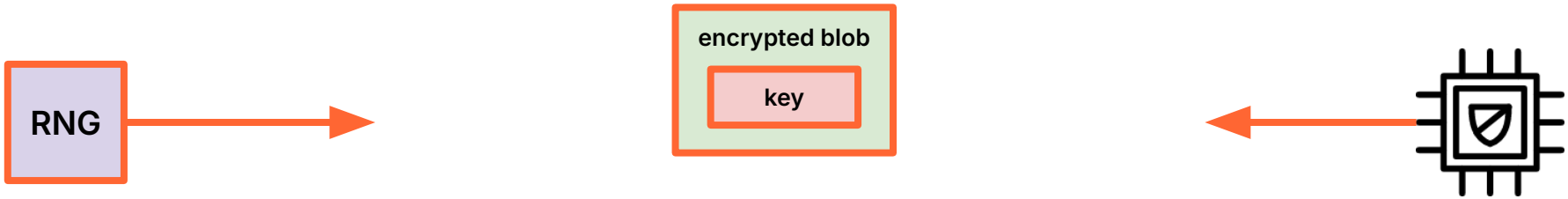
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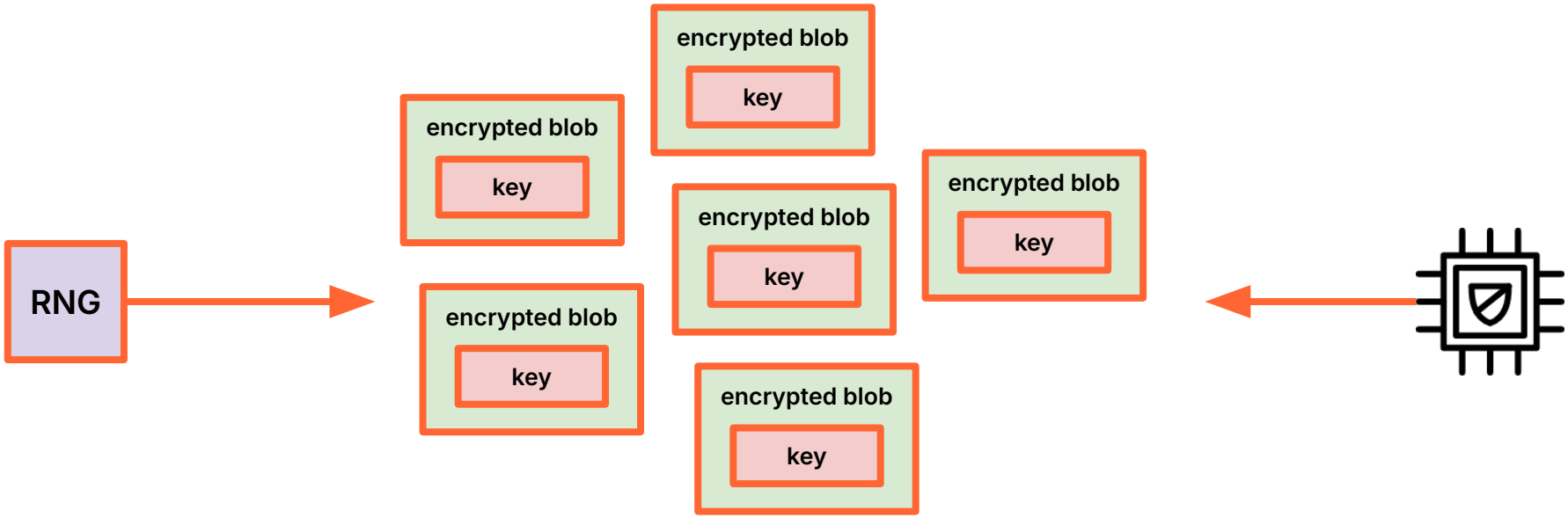
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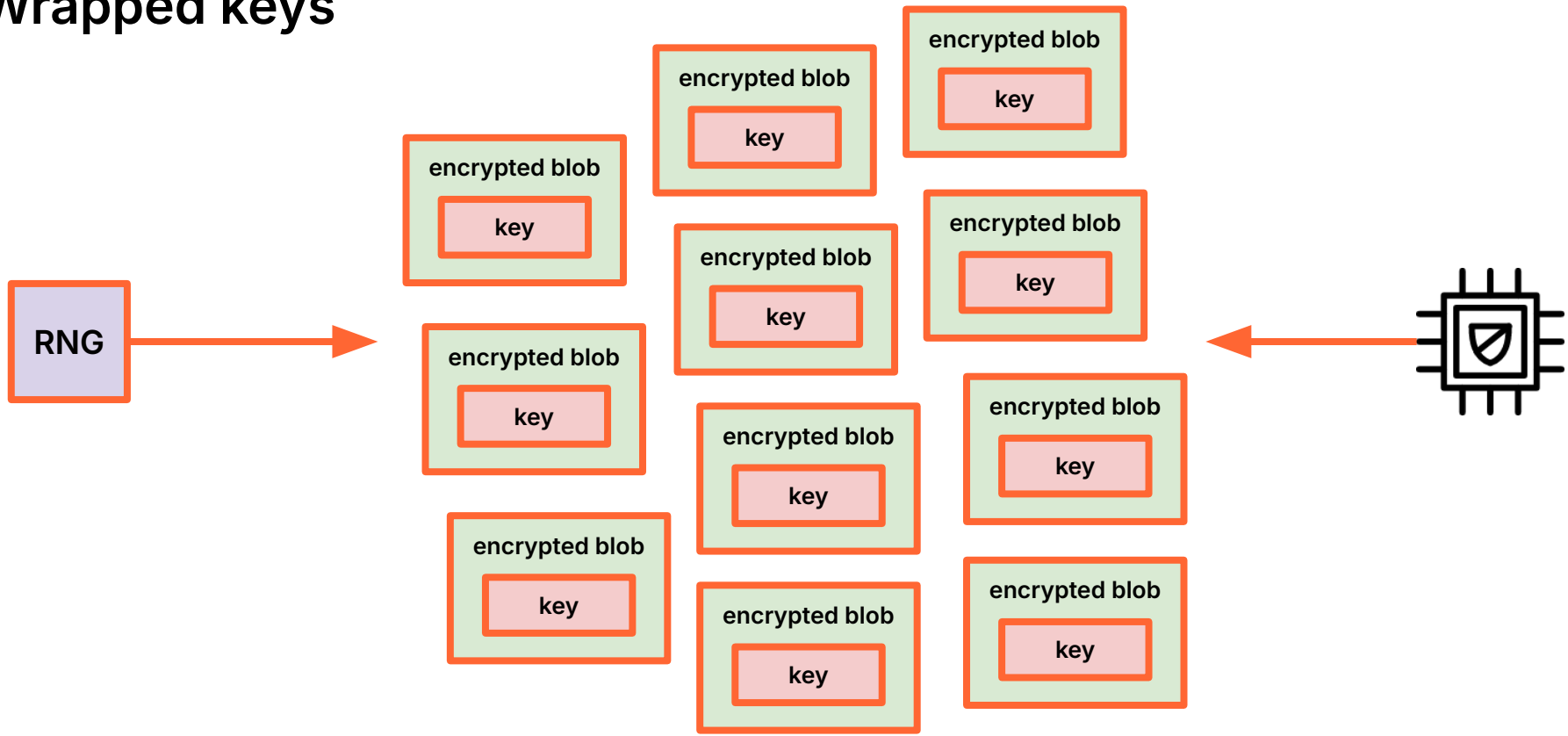
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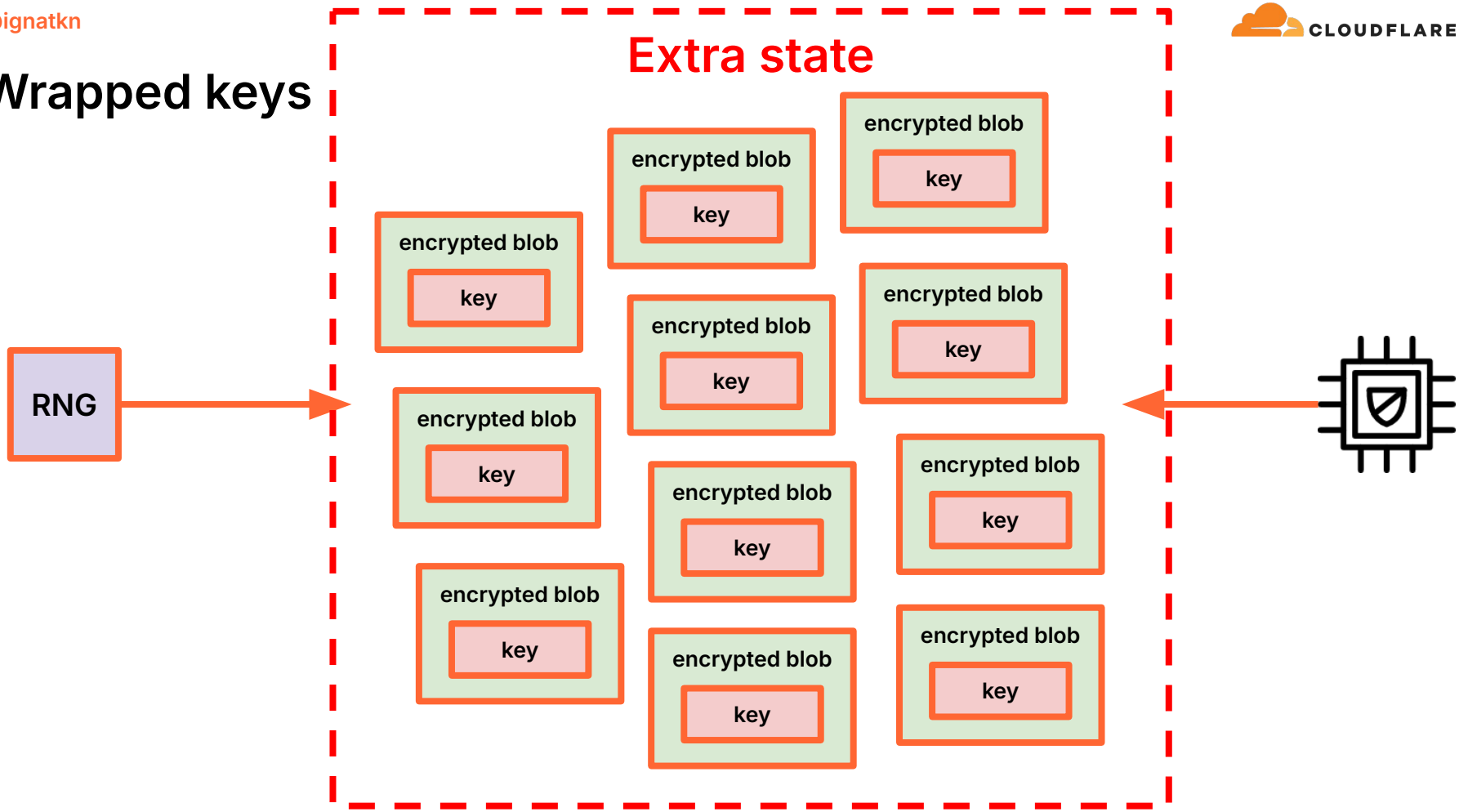
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Wrapped keys



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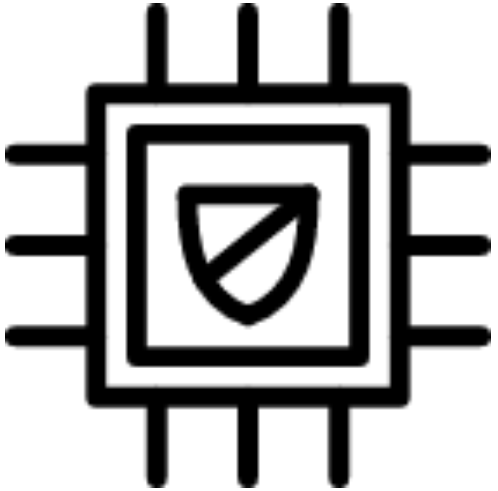
Wrapped keys



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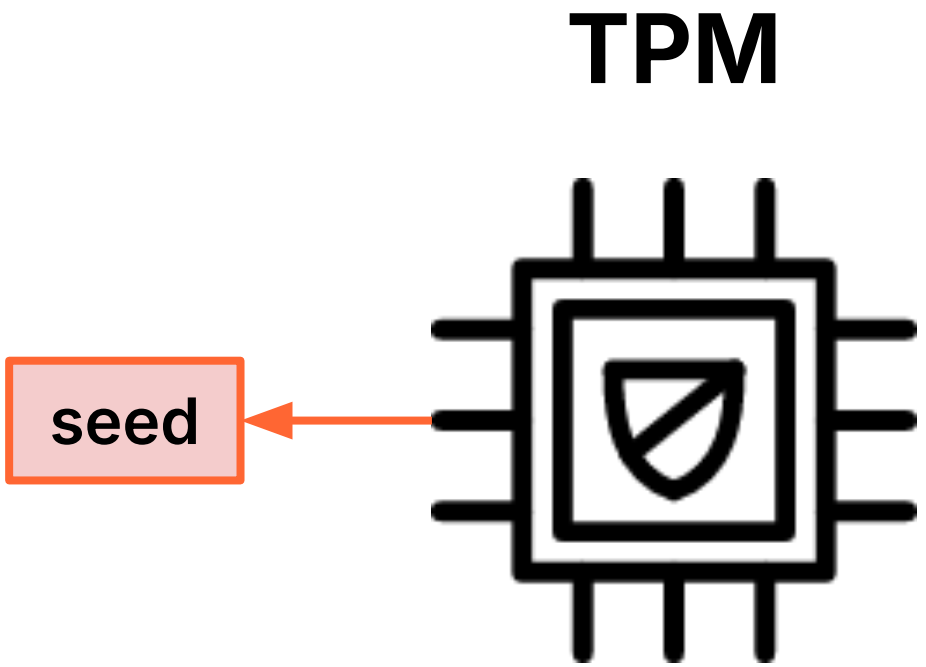
Derived keys

TPM



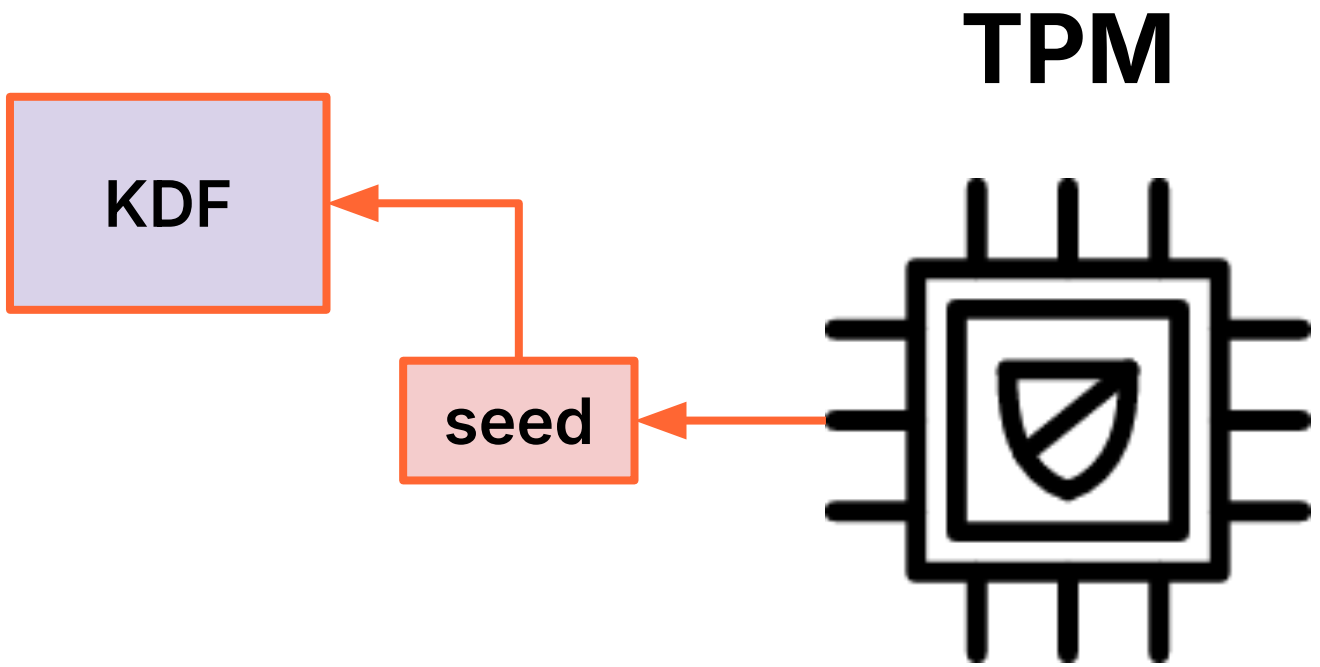
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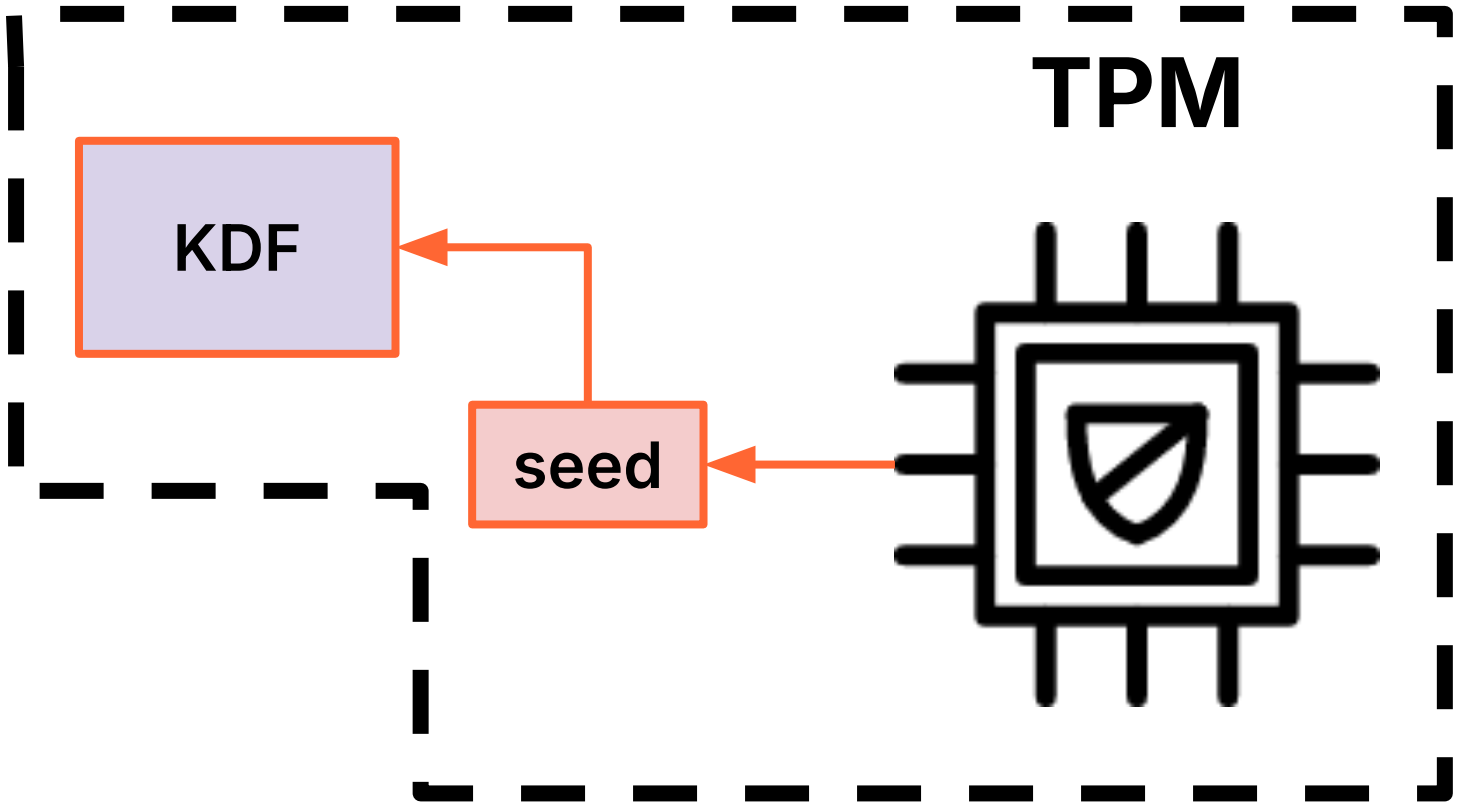
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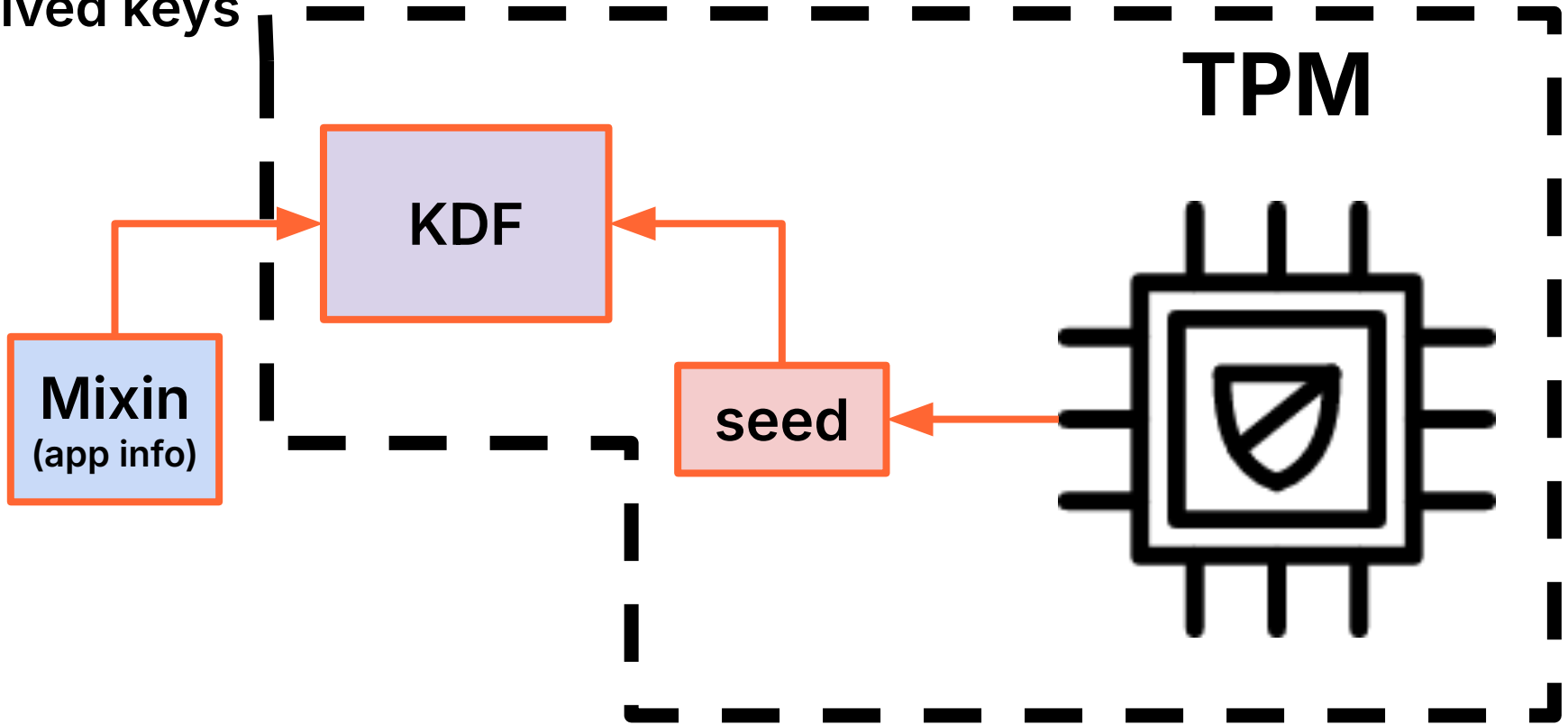
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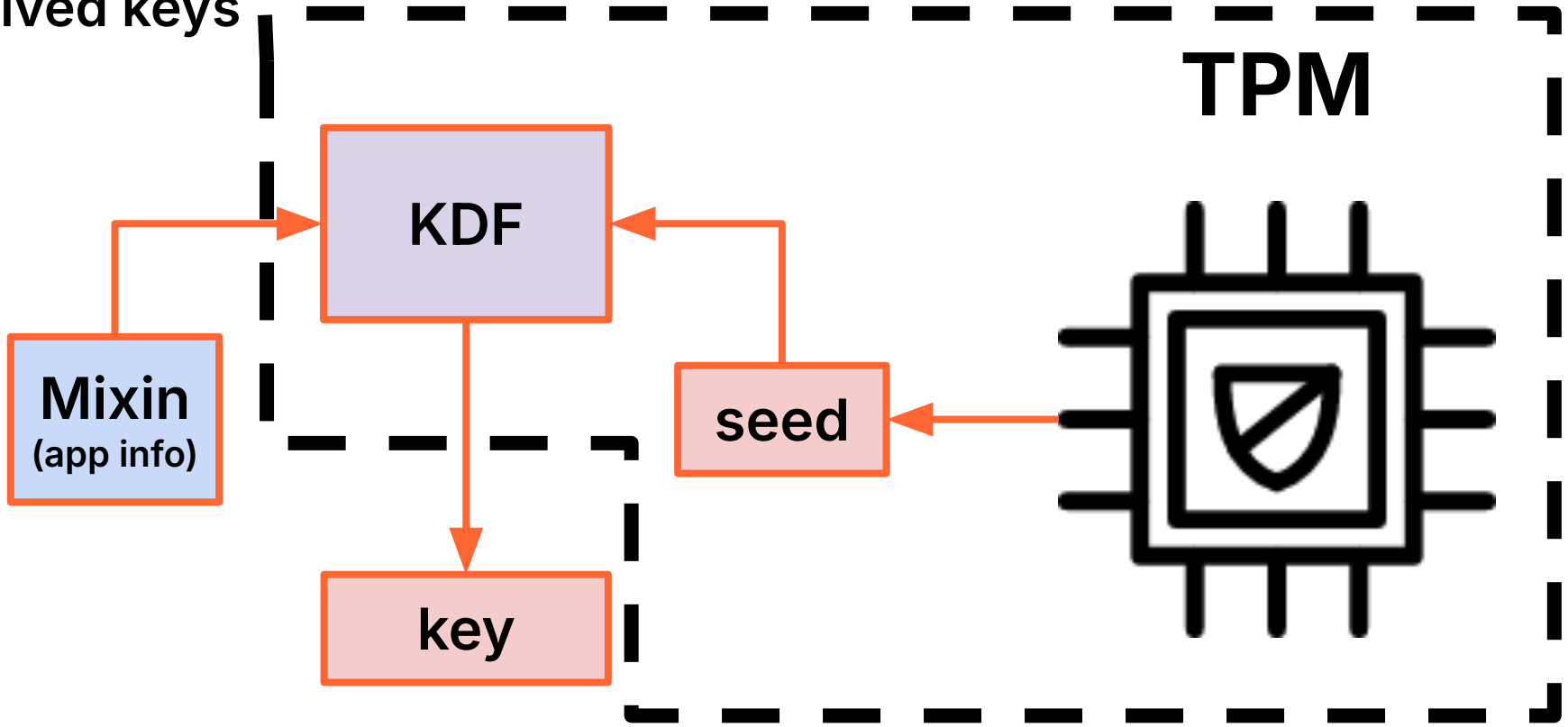
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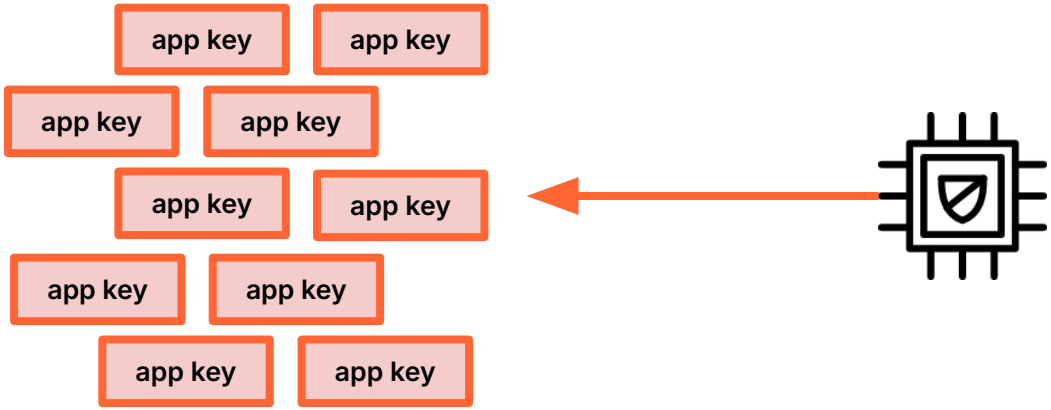
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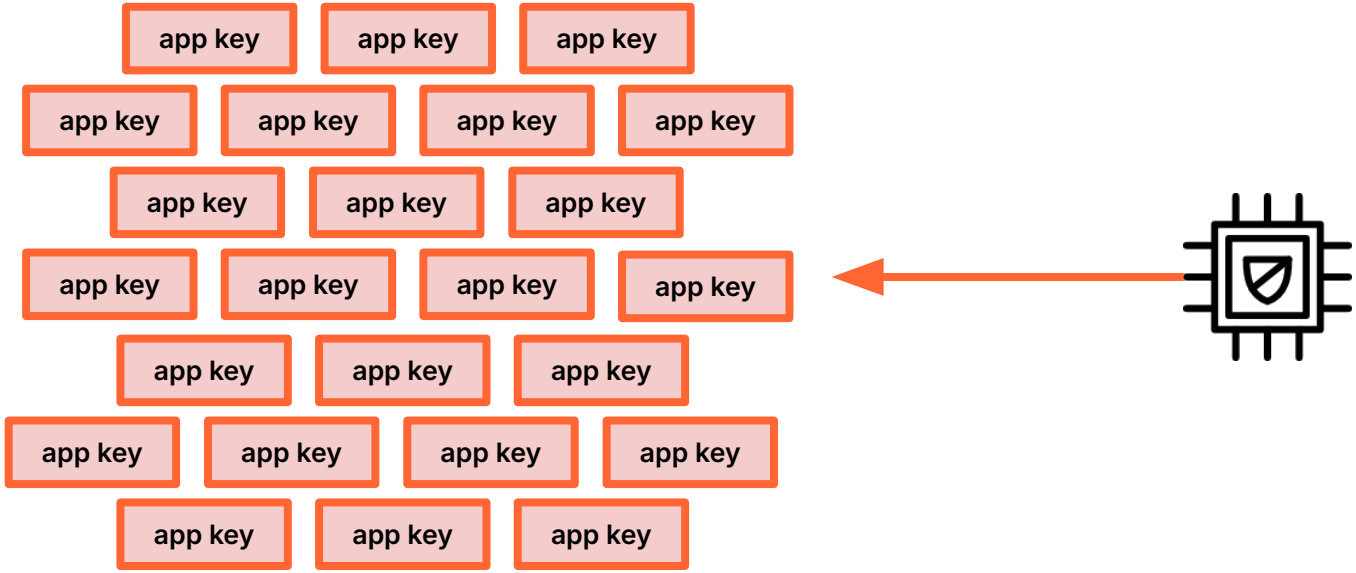
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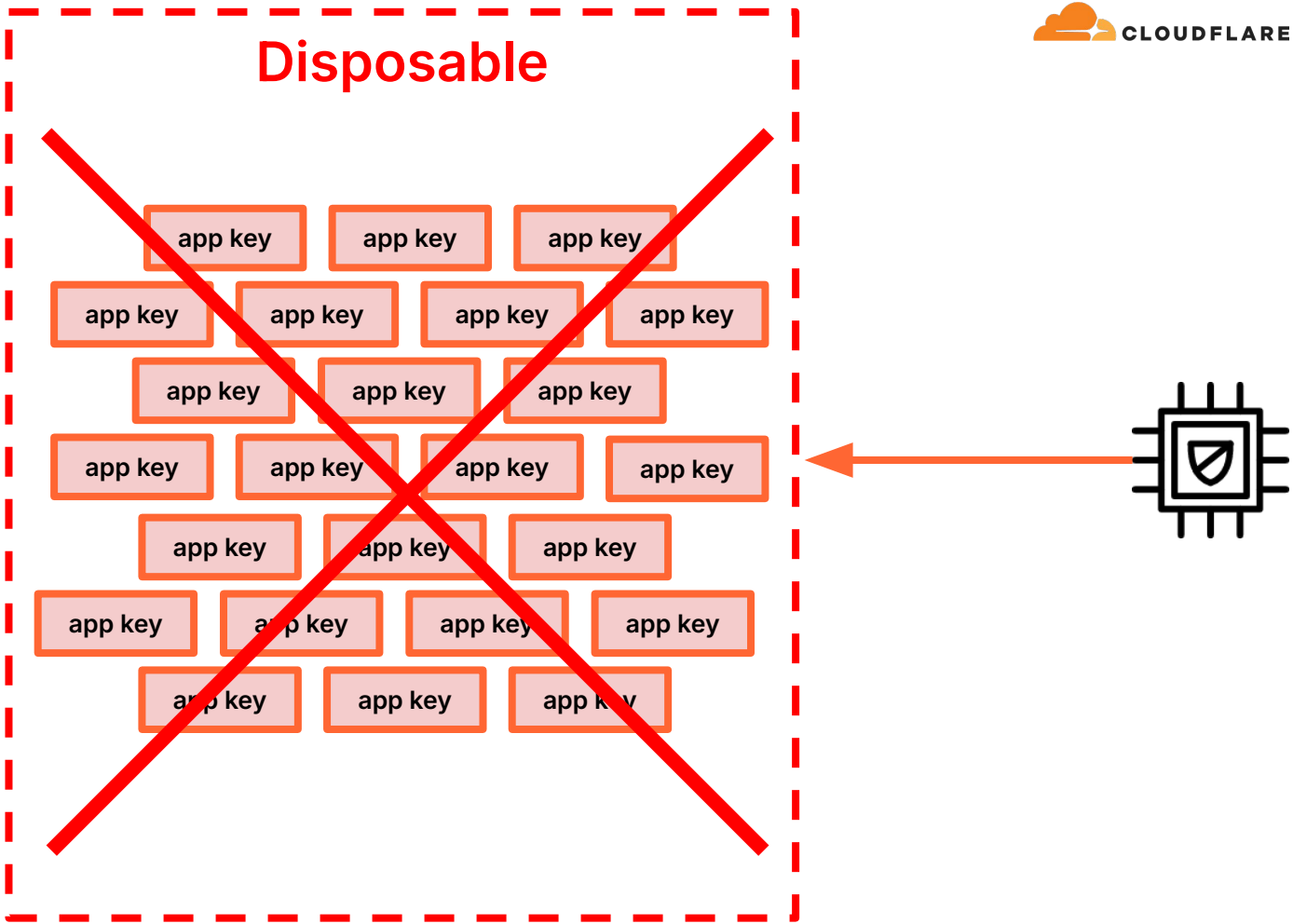
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Derived keys



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Derived keys



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TPM derived keys

- https://lore.kernel.org/all/20240503221634.44274-1-ignat@cloudflare.com/T/

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TPM derived keys

- https://lore.kernel.org/all/20240503221634.44274-1-ignat@cloudflare.com/T/
  - “I don't honestly believe that this will ever be a solution for any possible problem that exist in this planet”

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  - https://github.com/cloudflare/gokey

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TPM derived keys

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  - “I don't honestly believe that this will ever be a solution for any possible problem that exist in this planet”
  - https://optee.readthedocs.io/en/latest/architecture/porting_guidelines.html#hardware-unique-key
  - https://github.com/cloudflare/gokey
  - https://youtu.be/2RPcIbP2xsM?si=nyBCFkhuzwbXBLZf

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TPM derived keys

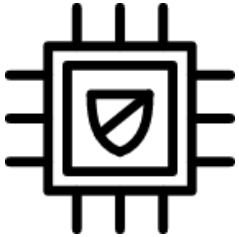
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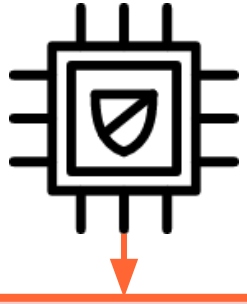
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  - https://youtu.be/2RPcIbP2xsM?si=nyBCFkhuzwbXBLZf
- Since then
  - Reimplemented key derivation approach: replaced TPM-based HMAC with TPM-based AES256CTR PRNG
  - Integrated support for TPM encrypted sessions
  - Compiles as a module

TPM derived keys

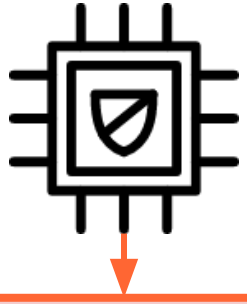


TPM derived keys



TPM2_CreatePrimary
(AES256CTR, UNIQUE)

TPM derived keys

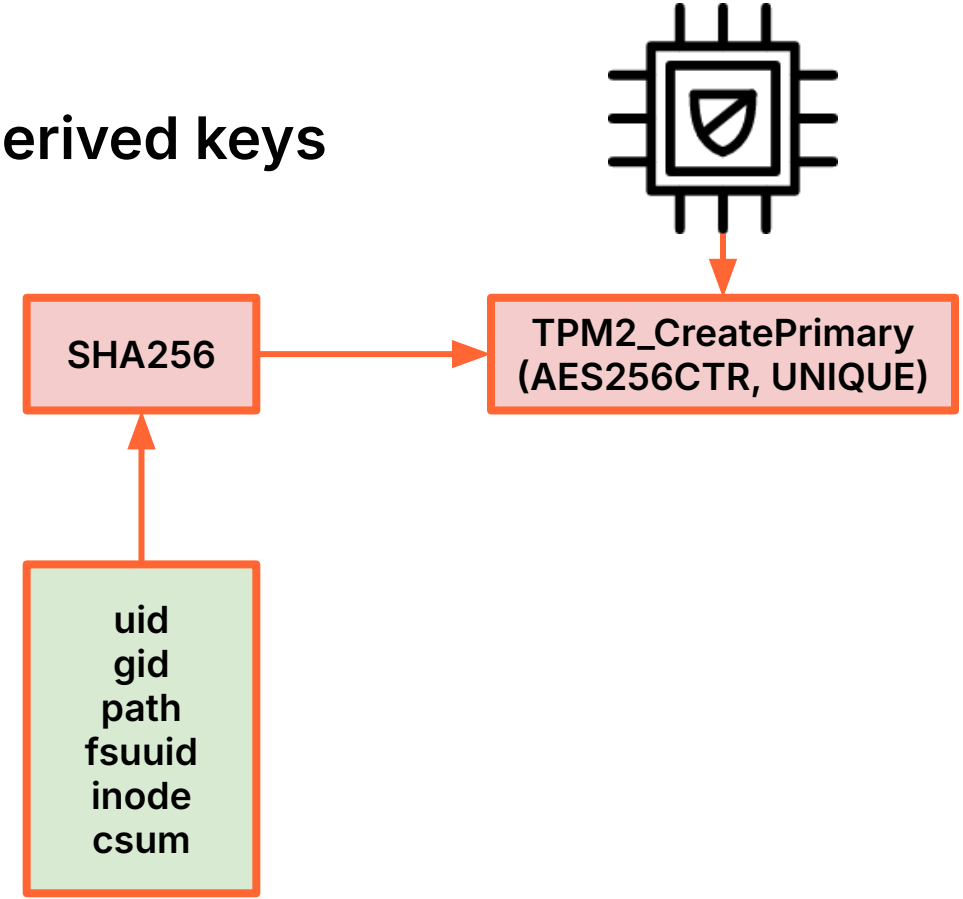


TPM2_CreatePrimary
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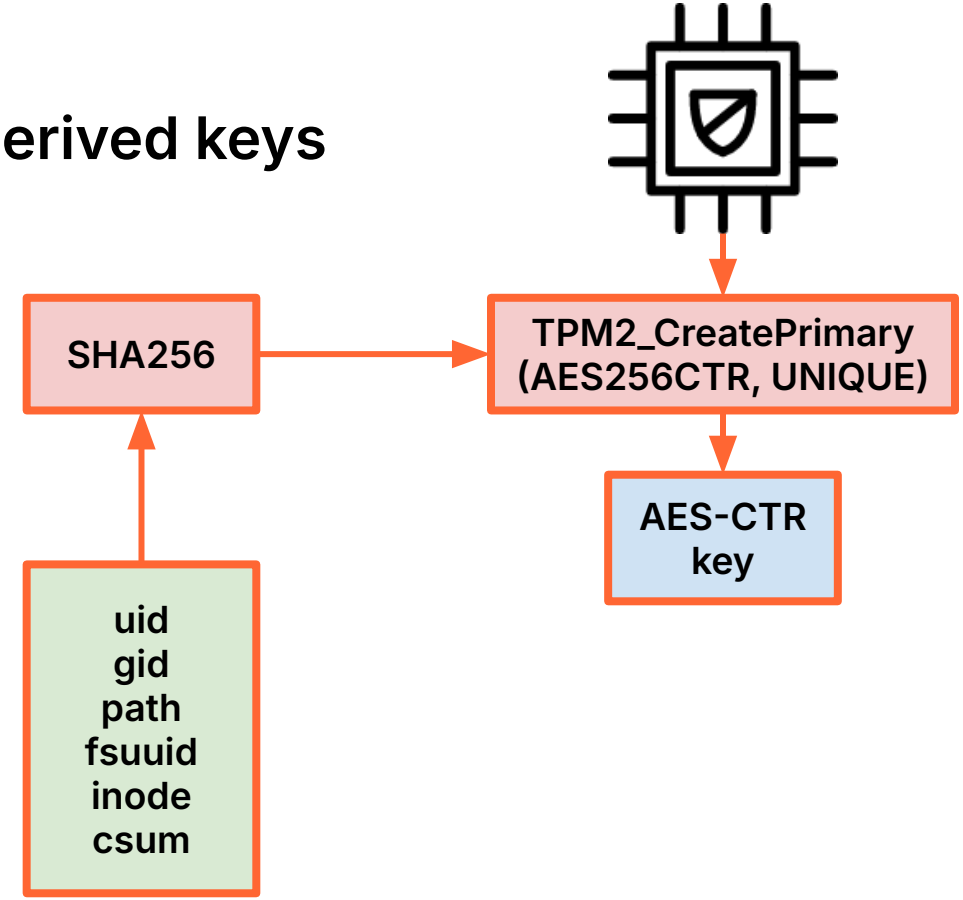
uid
gid
path
fsuuid
inode
csum

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TPM derived keys

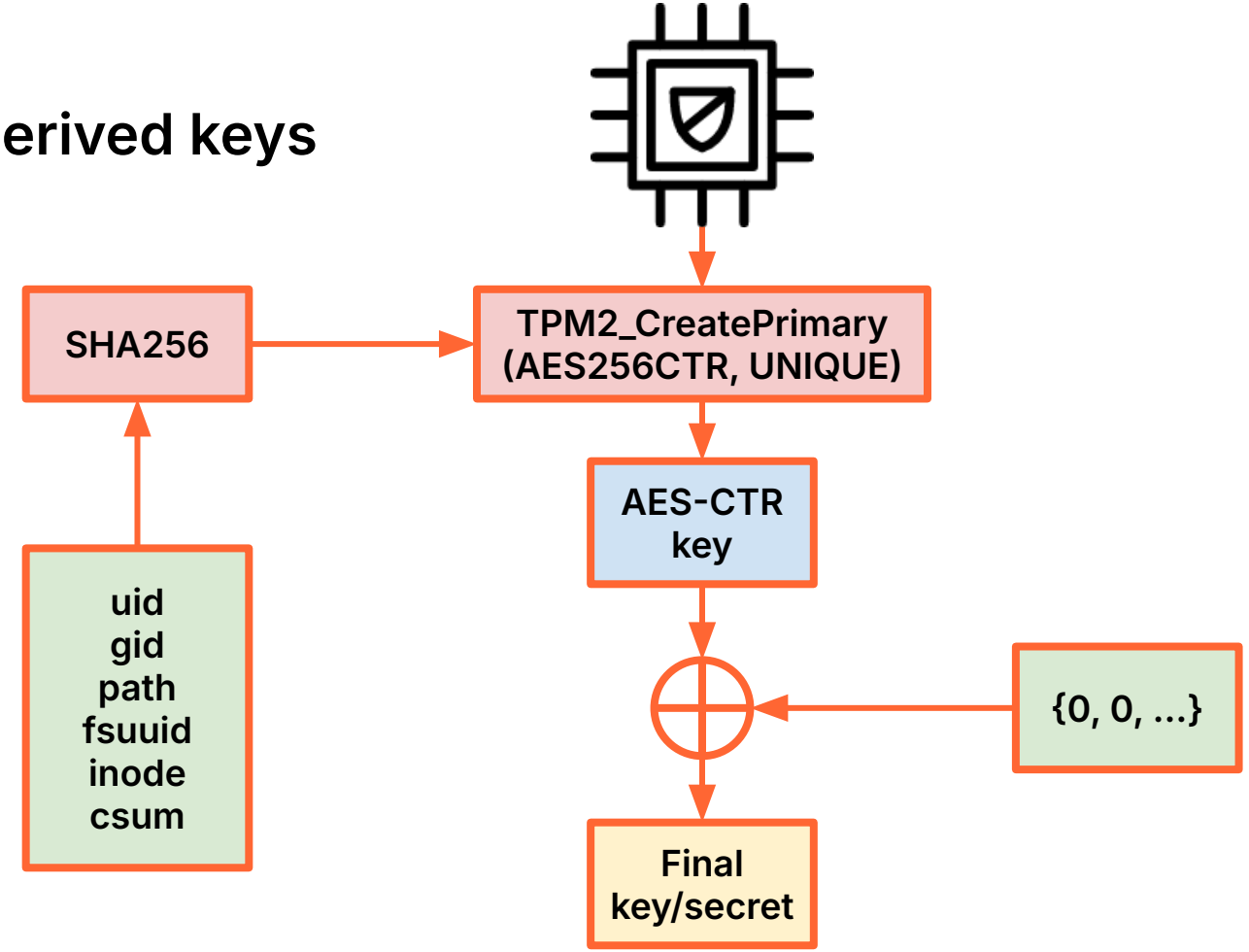


TPM derived keys



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TPM derived keys



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TPM user specific derived keys

```
$ keyctl request2 derived usr 'size=32 uid' @u
54647512
$ keyctl print 54647512
:hex:3254ee2cc1e1d79ccb22f15758a7b9504087222dd889a5efbc1c8b8b87b7148f
```

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TPM user specific derived keys

```
$ keyctl request2 derived usr 'size=32 uid' @u
54647512
$ keyctl print 54647512
:hex:3254ee2cc1e1d79ccb22f15758a7b9504087222dd889a5efbc1c8b8b87b7148f
$ keyctl unlink 54647512
1 links removed
```

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```
$ keyctl request2 derived usr 'size=32 uid' @u
54647512
$ keyctl print 54647512
:hex:3254ee2cc1e1d79ccb22f15758a7b9504087222dd889a5efbc1c8b8b87b7148f
$ keyctl unlink 54647512
1 links removed
$ keyctl request2 derived usr 'size=32 uid' @u
265546475
$ keyctl print 265546475
:hex:3254ee2cc1e1d79ccb22f15758a7b9504087222dd889a5efbc1c8b8b87b7148f
```

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TPM user specific derived keys

```
$ keyctl request2 derived usr 'size=32 uid' @u
54647512
$ keyctl print 54647512
:hex:3254ee2cc1e1d79ccb22f15758a7b9504087222dd889a5efbc1c8b8b87b7148f
$ keyctl unlink 54647512
1 links removed
$ keyctl request2 derived usr 'size=32 uid' @u
265546475
$ keyctl print 265546475
:hex:3254ee2cc1e1d79ccb22f15758a7b9504087222dd889a5efbc1c8b8b87b7148f
$ sudo keyctl request2 derived usr 'size=32 uid' @u
136468149
$ sudo keyctl print 136468149
:hex:9a360182012557e5e777ff3cb41737dcdaeab5efc8f0edbb4dbd3eaee41a4b3a
```

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TPM path specific derived keys

```
$ /usr/bin/keyctl request2 derived path 'size=16 path' @u
452878500
$ keyctl print 452878500
:hex:59e4f47de768245442b259fe7c1d63a3
```

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TPM path specific derived keys

```
$ /usr/bin/keyctl request2 derived path 'size=16 path' @u
452878500
$ keyctl print 452878500
:hex:59e4f47de768245442b259fe7c1d63a3
$ sudo cp /usr/bin/keyctl /opt/bin/
```

CLOUDFLARE

TPM path specific derived keys

```
$ /usr/bin/keyctl request2 derived path 'size=16 path' @u
452878500
$ keyctl print 452878500
:hex:59e4f47de768245442b259fe7c1d63a3
$ sudo cp /usr/bin/keyctl /opt/bin/
$ /opt/bin/keyctl request2 derived path2 'size=16 path' @u
857975889
$ keyctl print 857975889
:hex:d2a11b4c7c26989f7f2b540cfaee3614
```

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TPM path specific derived keys

```
$ keyctl request2 derived usrpath 'size=16 uid path' @u
232879988
$ keyctl print 232879988
:hex:f3bf22ea6af497e6db8a607d92458db8
```

CLOUDFLARE

TPM path specific derived keys

```
$ keyctl request2 derived usrpath 'size=16 uid path' @u
232879988
$ keyctl print 232879988
:hex:f3bf22ea6af497e6db8a607d92458db8
$ keyctl request2 derived fsinfo 'size=16 uid fsuuid inode' @u
521409059
$ keyctl print 521409059
:hex:cf1a11a887a5e09d8b04efac4233c32d
```

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TPM executable specific derived keys

```
$ /opt/bin/keyctl request2 derived exec 'size=16 csum' @u
92734356
$ keyctl print 92734356
:hex:904efb23fa808d9181a7824d1a251f41
```

CLOUDFLARE

TPM executable specific derived keys

```
$ /opt/bin/keyctl request2 derived exec 'size=16 csum' @u
92734356
$ keyctl print 92734356
:hex:904efb23fa808d9181a7824d1a251f41
$ sudo sed -i 's/Bad message/Bad massage/' /opt/bin/keyctl
$ keyctl unlink 92734356
1 links removed
```

CLOUDFLARE

TPM executable specific derived keys

```
$ /opt/bin/keyctl request2 derived exec 'size=16 csum' @u
92734356
$ keyctl print 92734356
:hex:904efb23fa808d9181a7824d1a251f41
$ sudo sed -i 's/Bad message/Bad massage/' /opt/bin/keyctl
$ keyctl unlink 92734356
1 links removed
$ /opt/bin/keyctl request2 derived exec 'size=16 csum' @u
251529703
$ keyctl print 251529703
:hex:020ea5f206f93f4c5d44519192888ef1
```

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“

What if the application needs to rotate keys because of a suspected compromise?



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TPM derived keys with labels

```
$ keyctl request2 derived v1 'size=16 label=v1' @u
274539938
$ keyctl print 274539938
:hex:b318728ff0f37e68bbaa5086480c3e25
```

CLOUDFLARE

TPM derived keys with labels

```
$ keyctl request2 derived v1 'size=16 label=v1' @u
274539938
$ keyctl print 274539938
:hex:b318728ff0f37e68bbaa5086480c3e25
$ keyctl request2 derived v2 'size=16 label=v2' @u
17456321
$ keyctl print 17456321
:hex:1e290b243aa46a9d2020eb14273dd214
```

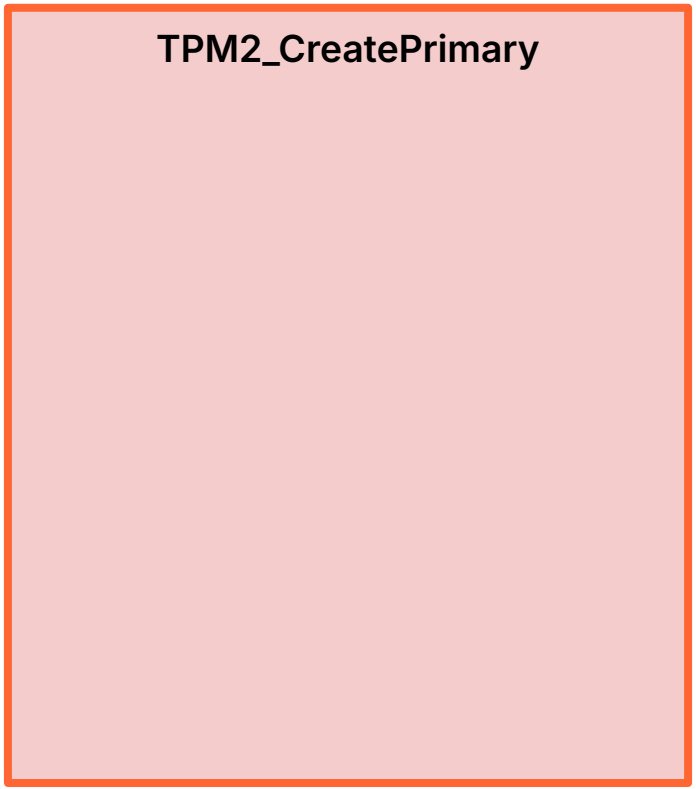
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> Those mixing values are well known ... then anyone with access to the TPM can derive the key from user space because they can easily obtain the mixing parameters and there's no protection to the TPM keyed hash operation



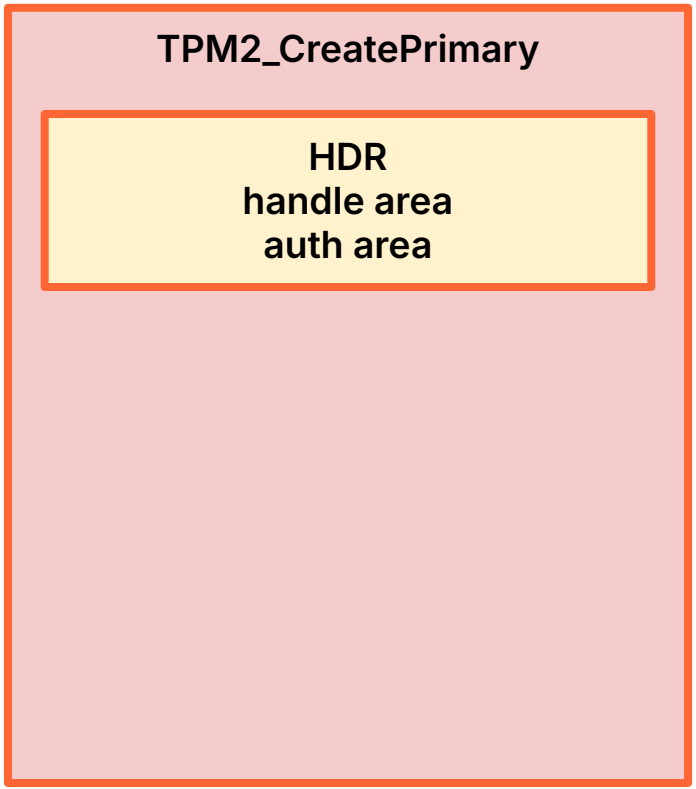
CLOUDFLARE

Kernel reserved unique domain



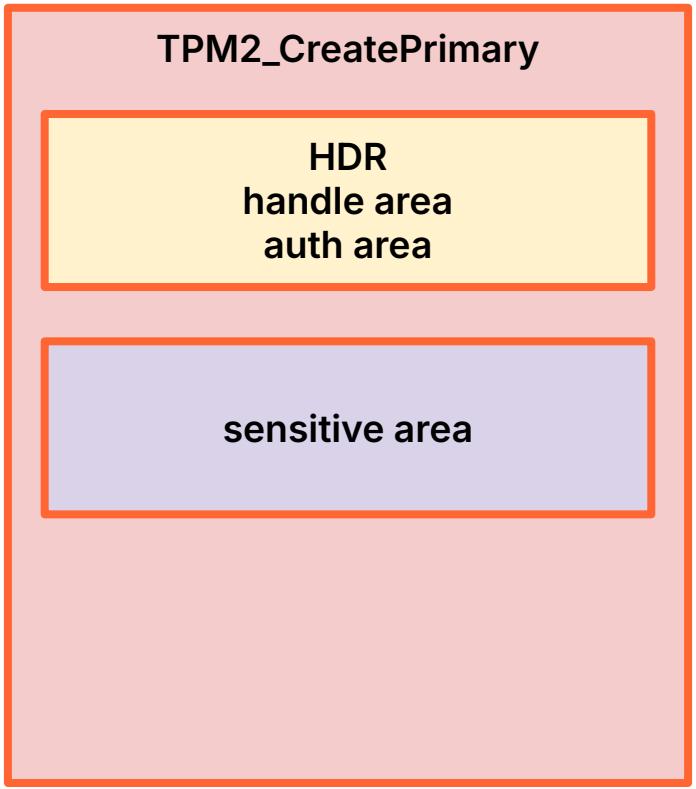
CLOUDFLARE

Kernel reserved unique domain



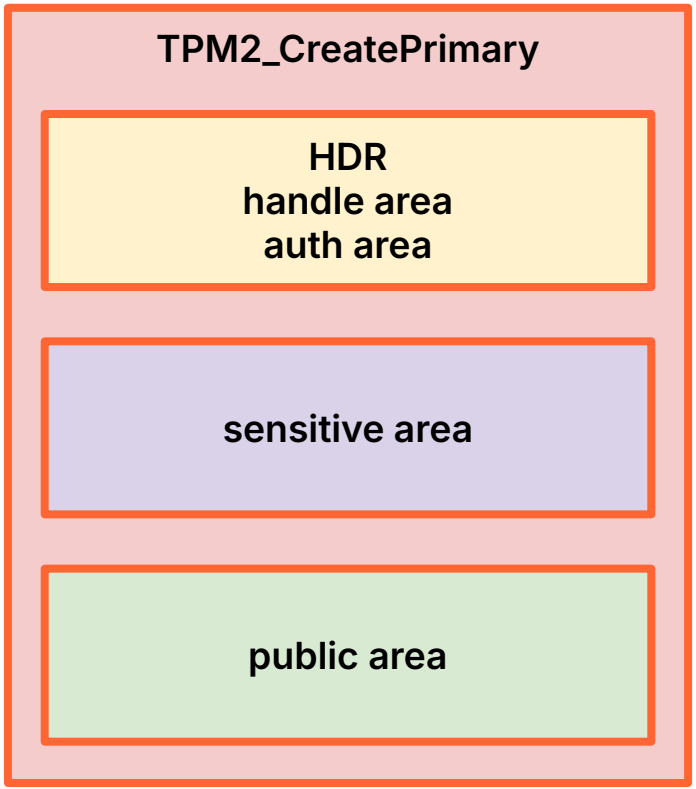
CLOUDFLARE

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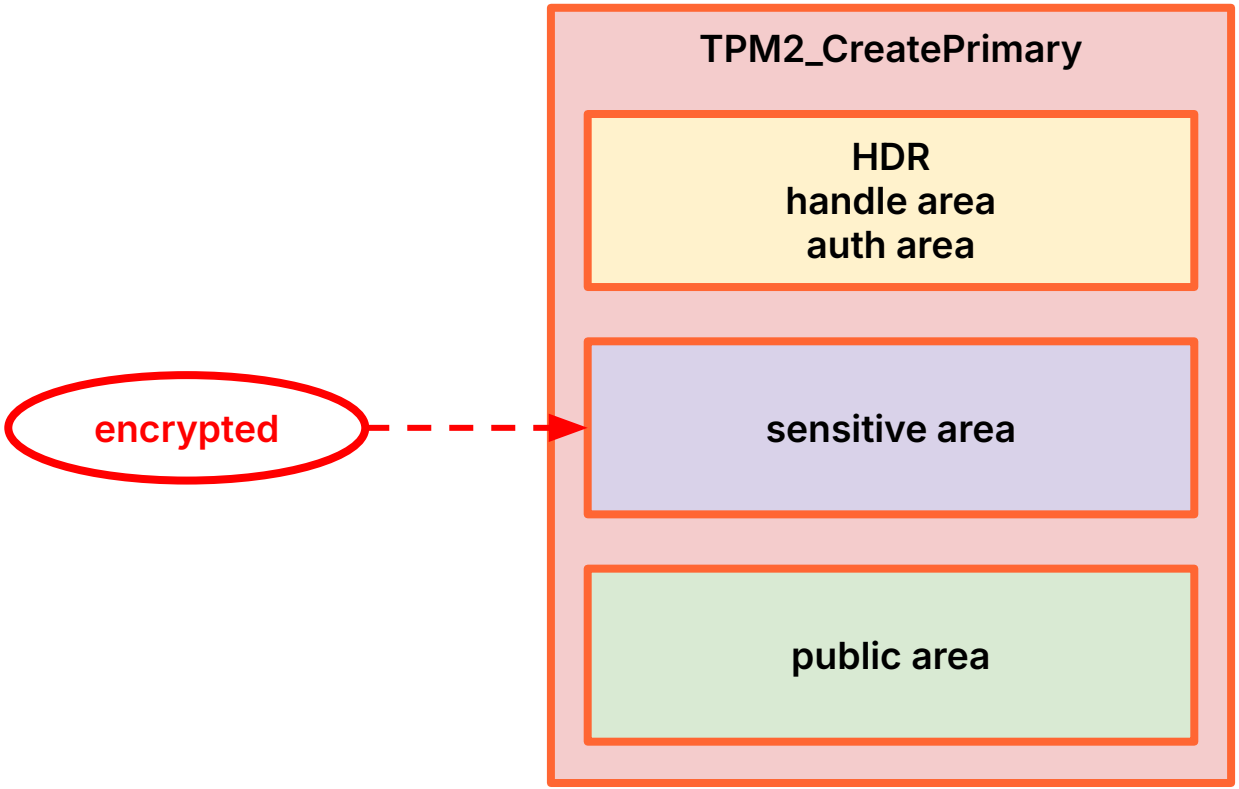
CLOUDFLARE

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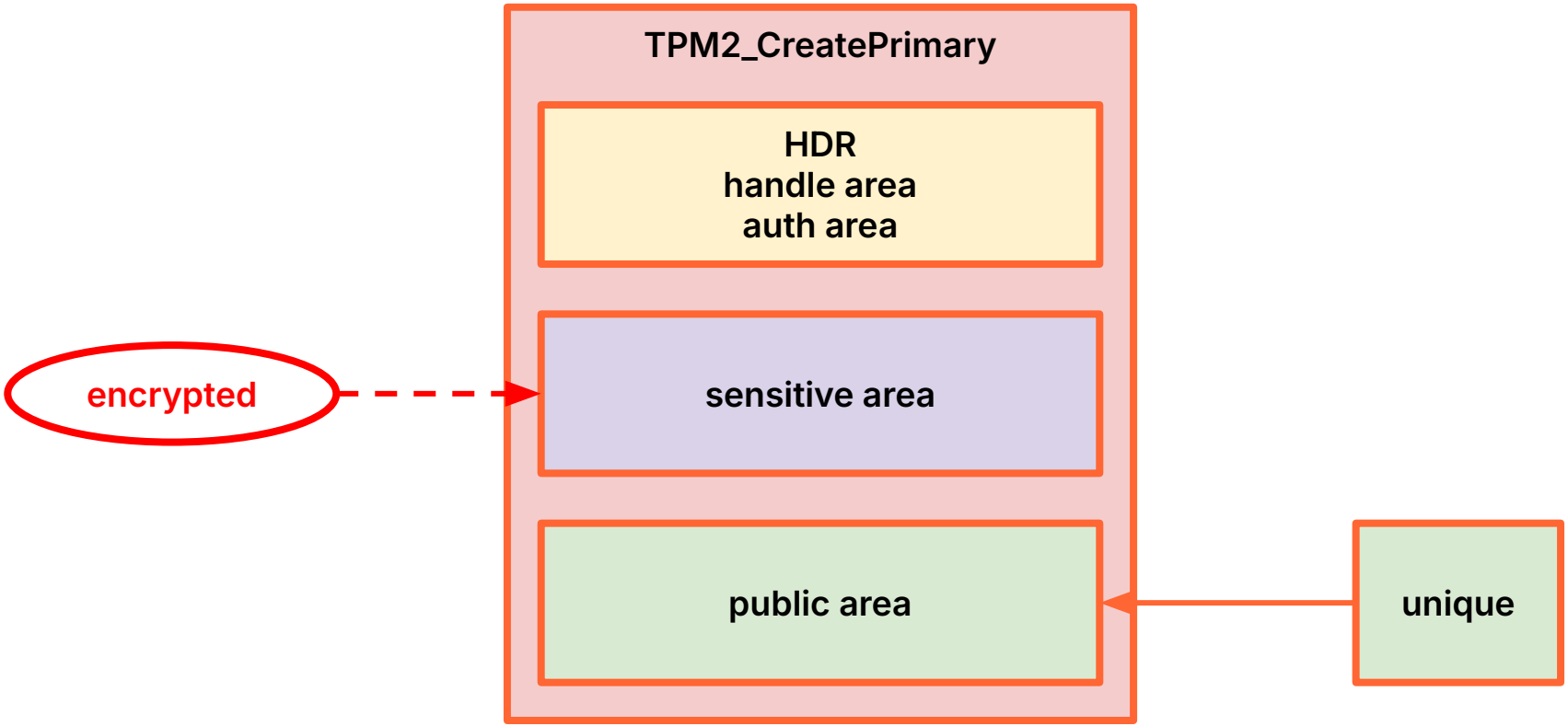
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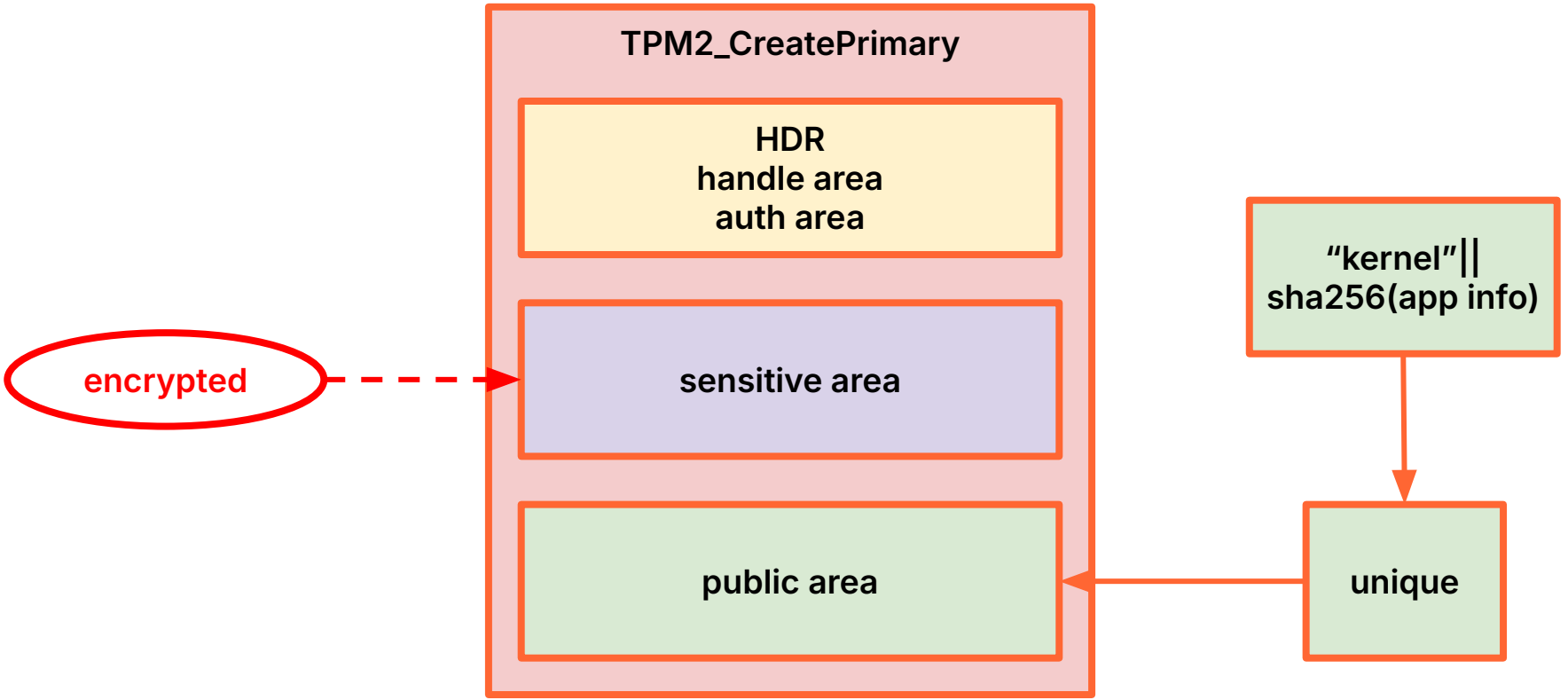
CLOUDFLARE

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```
$ echo abc | sudo tpm2_createprimary -G aes256ctr -u -
...
symcipher:
6d235c411a6f23ba2d88f81269cca65962a3a95b734ac9ba7d3fc6066
9b5fc24
```

CLOUDFLARE

Kernel reserved unique domain

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...
symcipher:
6d235c411a6f23ba2d88f81269cca65962a3a95b734ac9ba7d3fc6066
9b5fc24
$ echo kernel | sudo tpm2_createprimary -G aes256ctr -u -
ERROR:tcti:src/tss2-tcti/tcti-device.c:198:tcti_device_re
ceive() Failed to get response size fd 3, got errno 1:
Operation not permitted
...
```

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RFC

- What else is missing for easier adoption of TPMs in Linux software?
  - Any other application mixin parameters to be implemented?
  - Any other use-cases to be considered?

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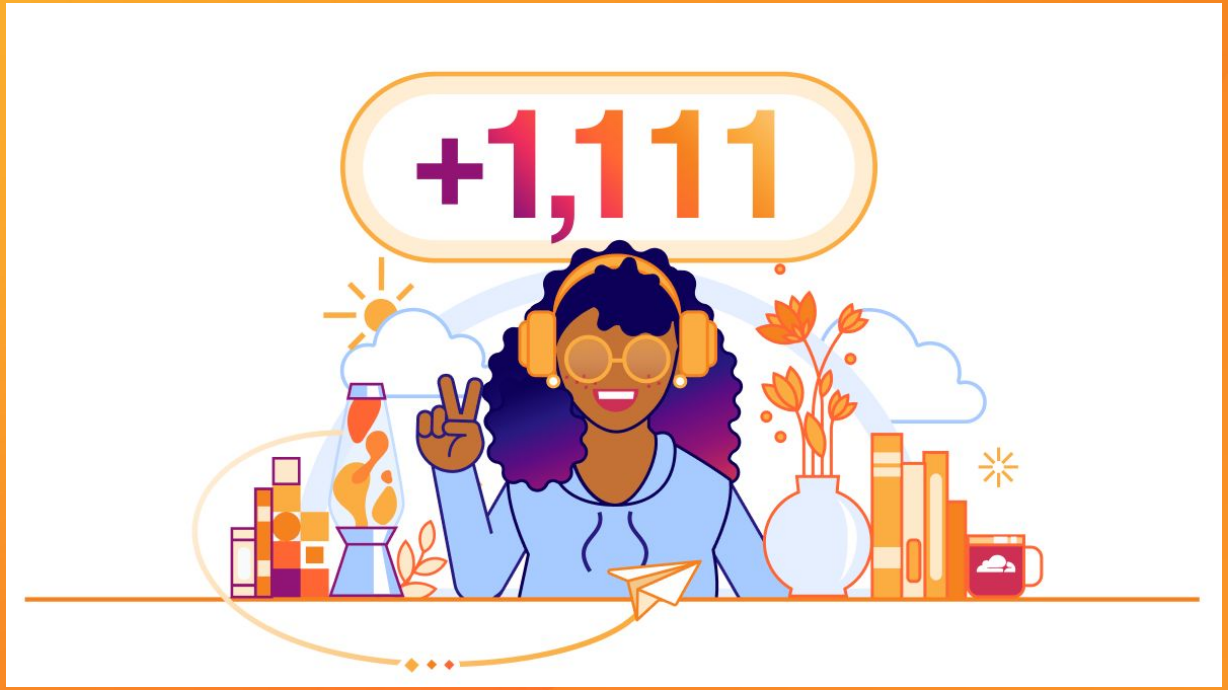
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https://github.com/ignatk/linux/tree/b4/tpm-derived-keys

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Thank you!



Questions?

https://blog.cloudflare.com/cloudflare-1111-intern-program/